

Solving Classical and Quantum Spin Glasses with Deep Boltzmann Quantum States

Luca Leone,^{*} Arka Dutta, and Markus Heyl

*Theoretical Physics III, Center for Electronic Correlations and Magnetism,
Institute of Physics, University of Augsburg, Augsburg, Germany.*

Enrico Prati and Pietro Torta

Department of Physics, University of Milan, Milan, Italy.

(Dated: October 3, 2025)

We introduce Deep Boltzmann Quantum States, a new quantum extension of deep Boltzmann machines. This architecture extends the physical Hilbert space by introducing layers of hidden quantum spins, hence inheriting efficient block Gibbs sampling and other computational advantages. By combining natural gradients and advanced stochastic optimizers, we successfully obtain the ground state of classical and quantum spin glasses with long-range interactions. When targeting hard ground states of frustrated models, we demonstrate that it is convenient to tune the hardness of the problem Hamiltonian by interpolating from an easy to a hard regime. Hence, we devise Neural Quantum Annealing, a hybrid algorithm that merges the principles of quantum annealing with neural-network wavefunctions to address large-scale optimization. By approximating the adiabatic dynamics on the variational manifold of Deep Boltzmann Quantum States, we obtain exact ground-state solutions of classical and quantum Ising spin glasses with hundreds of spins. Finally, we tackle a real-world NP-complete task known as the Job Shop Scheduling Problem and solve instances that exceed the current limitations of quantum annealing hardware. These findings highlight the effectiveness of integrating quantum annealing principles with the expressive power of modern neural architectures, providing a scalable framework for addressing real-world QUBO problems.

I. INTRODUCTION

Spin glasses are of central interest in many areas of science and technology: other than their relevance in describing the physics of disordered magnets [? ?], they are equally influential beyond physics as finding their ground states corresponds to solving many NP-hard optimization problems [? ?]. Consequently, spin glass ground-state problems have been the subject of extensive studies by many different scientific communities.

Quantum annealing (QA) provides a promising route to tackle these problems by leveraging the principles of adiabatic quantum computation (AQC). However, practical implementations of QA, such as D-Wave quantum annealers [1], face several limitations, including noise, thermal fluctuations, limited qubit connectivity, and scalability constraints, which impede their ability to handle large and complex problems.

In recent years, Neural Quantum States (NQS) have emerged as a powerful framework for representing quantum many-body wavefunctions, originally introduced in Ref. [2]. By leveraging the high expressivity of Artificial Neural Networks (ANNs), combined with the Stochastic Reconfiguration (SR) algorithm, NQS have been successfully used to solve a variety of ground-state problems [? ? ?]. Yet, a straightforward implementation of the NQS paradigm relies on Markov Chain Monte Carlo (MCMC) sampling, and is then not effective for spin glasses.

Training an NQS to solve spin glasses via simulated QA was proposed in Ref. [3] which introduced a variational

version of quantum annealing, Variational Quantum Annealing (VQA). VQA with Recurrent Neural Networks appeared less effective than its classical counterpart, and VQA results were only shown for polynomial complexity problems, such as the random Ising chain and the 2D Edwards–Anderson model without longitudinal field.

In this work, we introduce Boltzmann Quantum States (BQS), a class of variational Ansatz inspired by Boltzmann Machines (BMs). Starting from a physical system composed of N_v quantum spin degrees of freedom (the visible spins), we introduce an additional set of N_h hidden quantum spin variables that can be interpreted as unobserved degrees of freedom that yield no energy contribution to the system. Then, we use a Jastrow wavefunction [?] to approximate the quantum state of the extended system. As the hidden spins yield no energy contribution, they can be used to model higher-order interactions between the visible spins that are encoded in the entanglement between the visible and hidden spins. Finally, as in deep Boltzmann machines (DBMs), we impose limited inter-layer connectivity and thereby construct Deep Boltzmann Quantum States (DBQS). This construction preserves most of the properties and computational advantages of DBMs, and crucially, the possibility of efficient sampling via block Gibbs chains. We then introduce Neural Quantum Annealing (NQA), a hybrid algorithm to find the ground-state of a target Hamiltonian combining the ideas of VQA and SR with the expressivity and sampling efficiency of DBQS.

We evaluate NQA on classical and quantum spin glasses. For Ising encodings of Job Shop Scheduling Problem (JSSP) instances, we obtain exact ground states up to 725 spins that exceed the limitations of state of the

^{*} Also at Department of Physics, University of Milan, Milan, Italy.

art quantum annealing hardware, and for the Sherrington–Kirkpatrick (SK) model we reach exact or near-exact solutions up to 200 spins. We further find evidence consistent with ground state via energy–variance extrapolation for instances of the transverse-field SK model with 100 spins deep within the spin-glass phase.

Overall, this work demonstrates that combining deep neural-network Ansätze with annealing-inspired algorithms provides a powerful and flexible framework for addressing spin-glass physics and combinatorial optimization problems at scales beyond the reach of existing quantum annealing hardware and classical methods.

A python implementation of NQA based on JAX and conveniently packaged in an easy to deploy nqa-server similar to the spin glass server [4] is freely available on github.

II. NEURAL QUANTUM STATES

Neural Quantum States (NQS) leverage the variational principle, combined with the expressive power of Artificial Neural Networks (ANNs). In this framework, the wavefunction of a quantum many-body system is represented by a parameterized ANN, removing the need for an ad-hoc variational Ansatz based on physical intuition. Originally introduced in Ref. [2], NQS have been successfully applied to solve a variety of quantum ground-states [5–8] and there are also promising results for the study of quantum dynamics [2, 9–12]. This section provides the basic background on NQS for spin-1/2 systems necessary for the remainder of this article. For a comprehensive review, see e.g. Ref. [13, 14].

Formally, we consider a system of N spin-1/2 degrees of freedom, whose basis states are labeled by spin configurations $\mathbf{x} = (x_1, x_2, \dots, x_N)$, with $x_i = \pm 1$. This basis is often conventionally associated with the eigenbasis of Pauli-Z operators¹, and the general many-body wavefunction can be expanded as

$$|\psi\rangle = \sum_{\mathbf{x}} \psi(\mathbf{x}) |\mathbf{x}\rangle. \quad (1)$$

In the NQS framework, the wavefunction is expressed as

$$\psi(\mathbf{x}; \boldsymbol{\theta}) \equiv \langle \mathbf{x} | \psi(\boldsymbol{\theta}) \rangle = f_{\text{ANN}}(\mathbf{x}; \boldsymbol{\theta}), \quad (2)$$

where $f_{\text{ANN}}(\mathbf{x}; \boldsymbol{\theta})$ is the output of an ANN that takes $\mathbf{x}$ as an input, and $\boldsymbol{\theta}$ is its set of trainable complex or real parameters.

Leveraging the Time-Dependent Variational Principle (TDVP), it is possible to map the dynamics of the quantum state to a system of ordinary differential equations

for the variational parameters [2],

$$i |\dot{\psi}(\boldsymbol{\theta})\rangle = H |\psi(\boldsymbol{\theta})\rangle \iff \dot{\boldsymbol{\theta}} = -i S^{-1}(\boldsymbol{\theta}) F(\boldsymbol{\theta}). \quad (3)$$

Here, H denotes the system Hamiltonian, S is the Fisher information matrix (also known as the quantum geometric tensor), and F is the force array, defined as

$$\begin{cases} S_{kl}(\boldsymbol{\theta}) = \langle O_k \bar{O}_l \rangle_p - \langle O_k \rangle_p \langle \bar{O}_l \rangle_p, \\ F_k(\boldsymbol{\theta}) = \langle E_{\text{loc}} \bar{O}_k \rangle_p - \langle E_{\text{loc}} \rangle_p \langle \bar{O}_k \rangle_p. \end{cases} \quad (4)$$

with O_k and E_{loc} representing, respectively, the variational derivatives and the local energy

$$\begin{aligned} O_k(\mathbf{x}; \boldsymbol{\theta}) &= \frac{d}{d\theta_k} \log \psi(\mathbf{x}; \boldsymbol{\theta}), \\ E_{\text{loc}}(\mathbf{x}; \boldsymbol{\theta}) &= \frac{\langle \mathbf{x} | H | \psi(\boldsymbol{\theta}) \rangle}{\psi(\mathbf{x}; \boldsymbol{\theta})}. \end{aligned} \quad (5)$$

In Eq. (4) we indicated with $\langle \cdot \rangle_p$ the expectation value over the probability distribution

$$p(\mathbf{x}; \boldsymbol{\theta}) = \frac{|\psi(\mathbf{x}; \boldsymbol{\theta})|^2}{\langle \psi(\boldsymbol{\theta}) | \psi(\boldsymbol{\theta}) \rangle} = \frac{|\psi(\mathbf{x}; \boldsymbol{\theta})|^2}{\sum_{\mathbf{x}'} |\psi(\mathbf{x}'; \boldsymbol{\theta})|^2}. \quad (6)$$

Since the analytical computation of expectation values on an NQS is generally intractable, practical implementations of the TDVP rely on Monte Carlo sampling to obtain empirical estimates of the quantities in Eq. (4). More generally, the expectation value of an operator O can be estimated by computing the empirical average over the Born probability of

$$O_{\text{loc}}(\mathbf{x}; \boldsymbol{\theta}) = \frac{\langle \mathbf{x} | O | \psi(\boldsymbol{\theta}) \rangle}{\langle \mathbf{x} | \psi(\boldsymbol{\theta}) \rangle}, \quad (7)$$

which is known as the local operator (or local estimator).

Finally, NQS can be employed in the context of ground-state search by carrying out imaginary time evolution. This is formally done by redefining $\tau = it$ in Eq. (3). The intuition behind this approach is that the repeated application of the imaginary time evolution operator $U = e^{-\delta\tau H}$ for short imaginary time steps $\delta\tau$ to an initial state, exponentially suppresses higher-energy components. This results in effectively projecting the state onto the ground state, provided the initial state has a nonzero overlap with it. This approach is equivalent to minimizing the variational energy

$$E(\boldsymbol{\theta}) = \frac{\langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle}{\langle \psi(\boldsymbol{\theta}) | \psi(\boldsymbol{\theta}) \rangle} \quad (8)$$

via Natural Gradient Descent [15, 16], an algorithm known as Stochastic Reconfiguration (SR) [17] (see Appendix A).

In practice, the inversion of the Fisher information matrix S is required (as in Eq. (3)), which represents a computational bottleneck. Since S may not be invertible, the Moore-Penrose pseudo-inverse S^+ is often used instead [2, 7, 18], and a small diagonal shift can be applied

¹ We denote the Pauli-Z eigenvalues with $x_i = \pm 1$ for notational consistency with the following sections, since this is identified as the computational basis.

to the Fisher information matrix to stabilize the numerics. A variant of SR, known as min-SR, has been recently proposed in [7], often offering a significant computational speed-up by requiring the inversion of a smaller matrix. Since these two procedures are formally equivalent, the choice between them is guided only by computational convenience.

Solving the real-time TDVP dynamics is numerically challenging, primarily due to numerical instabilities arising from multiple sources [14]. A small integration time step δt and a large number of Monte Carlo samples are typically required to accurately solve for real-time dynamics. On the contrary, ground-state search with NQS is defined as the optimization problem in Eq. (8). When solved with SR, the objective is only to approximate the asymptotic state of the imaginary-time TDVP equations rather than to follow the complete imaginary-time trajectory. Hence, to solve the ground-state optimization problem, it is not strictly necessary to use standard SR, and natural gradients can be combined with advanced stochastic optimizers developed for deep learning, which can dramatically speed up convergence. In this work, we combine natural gradients with momentum [19–21], as discussed in Section VII. To simplify notation, we will use SR to denote the generalized scheme for optimizing an NQS in ground state search, which may combine standard SR (or min-SR) with advanced optimizers, as detailed in Appendix A.

III. ADIABATIC QUANTUM COMPUTATION AND PRACTICAL QUANTUM ANNEALING

Quantum Annealing (QA) [22–26] conceptually relies on the theory of Adiabatic Quantum Computation (AQC) [27]. In this framework, a quantum system is initialized in the easy-to-prepare ground state of a non-interacting *driver* Hamiltonian, often selected as the transverse-field Hamiltonian

$$H_0 = H_x = - \sum_{j=1}^N \sigma_j^x. \quad (9)$$

The system is then slowly driven out of equilibrium by tuning an external coupling $s = s(t)$ in an interpolating Hamiltonian

$$H(s) = s H_T + (1 - s) H_0. \quad (10)$$

The objective is to obtain the ground state of the *target* Hamiltonian H_T , which may either describe a quantum many-body system or encode the solution(s) of a hard classical computational problem in its lowest energy state(s). In an ideal AQC scenario, this is achieved by implementing adiabatic dynamics that, by definition, follow the instantaneous ground-state of $H(s)$, as the coupling $s(t)$ is slowly varied from $s(0) = 0$ to $s(\tau) = 1$ over a sufficiently long total annealing time τ . The system is implicitly assumed to undergo Schrödinger evolution,

corresponding to a closed quantum system at zero temperature.

When dealing with classical optimization, the target Hamiltonian is often in the form of an Ising spin glass:

$$H_T = H_z = \sum_{i=1}^N \sum_{j=i+1}^N J_{ij} \sigma_i^z \sigma_j^z + \sum_{i=1}^N h_i \sigma_i^z, \quad (11)$$

where the non-zero couplings J_{ij} define an interaction graph among the spin variables and the h_i are local longitudinal fields. This is equivalent to Quadratic Unconstrained Binary Optimization (QUBO) problems and is precisely the type of models that are natively supported on current superconducting QA hardware (e.g., D-Wave systems [1]). Several critical technological and scientific tasks can be reformulated in terms of QUBO [28] or, equivalently, as an Ising spin glass [29], as this represents the prototype of an NP-complete problem.

The theoretical AQC framework encounters severe practical limitations. To avoid exciting higher energy states during the dynamics, a large value of τ is theoretically required by the adiabatic theorem(s) [27]. This value critically depends on the minimum spectral gap Δ encountered during the annealing. In simplified terms, the minimum annealing time τ required for adiabaticity typically scales as Δ^{-2} [27]. However, since the annealing path connects a paramagnetic phase in the x direction to a disordered spin glass phase in the z direction, a first-order quantum phase transition is often present in the thermodynamic limit, where the spectral gap closes exponentially fast with the system size N [30]. Hence, the value of τ required for adiabaticity typically diverges exponentially in N , representing a major computational bottleneck for AQC [31–33].

While quantum annealers closely implement Eqs. (10)–(11), an estimate of the order of magnitude of τ required for adiabaticity is often unfeasible, and practical implementations are also affected by thermal noise, decoherence effects, and hardware constraints [34]. An additional challenge, related to limited hardware connectivity, is the necessity of performing an embedding of the interactions in Eq. (11) into the hardware graph [35]. In practice, current QA hardware does not implement AQC, but is rather employed as a heuristic sampler of the low-energy states of the Hamiltonian in Eq. (11). Drawing a sample requires system initialization, quantum evolution, and (destructive) measurement on the final state, and this whole process is repeated many times (typically thousands of shots). Hence, rather than a single zero-temperature adiabatic evolution, quantum annealers perform several shorter evolutions with a finite value of τ related to device constraints, each of which is often modeled as an open quantum system dynamics [27].

This heuristic approach shows promising results in several applications, especially when utilized as a subroutine in a hybrid quantum-classical algorithm [34, 36] or guided through learning procedures [37], yet no conclusive evidence of quantum speedup [38] to solve a class of models

in the category of Eq. (11) has been shown to date. Recent progress [39] has been demonstrated with D-Wave hardware in the regime of fast annealing (τ of the order of a few ns), where coherent Schrödinger dynamics can be realized to simulate the physics of quantum spin glasses on hardware-compatible geometries at the boundary or beyond classical simulations (see also Refs [40, 41] for improved classical simulations). This regime is opposite to the adiabatic theory, and the implications for solving hard *classical* optimization problems as in Eq. (11) remain unclear.

Finally, it is relevant to mention that the abstract framework of AQC can, in principle, be adapted to prepare the quantum ground state of any many-body target Hamiltonian. A straightforward example is represented by a quantum Ising spin glass, simply obtained as in Eqs. (9)–(11), by annealing to a final finite value of transverse field coupling, i.e. to $s(\tau) < 1$. Even more generally, various choices for H_0 and H_T can be envisaged, along with different schedules for the time-dependent couplings. Practical realizations of this broad computational scheme, known as *analog quantum computation*, have been implemented on different experimental platforms, such as trapped Rydberg atom arrays [42].

IV. VARIATIONAL CLASSICAL ALGORITHMS INSPIRED BY QUANTUM ANNEALING

The concepts and tools of AQC and QA have inspired different classical quantum-inspired variational algorithms that perform real and/or imaginary time evolution on a variational manifold [3, 43, 44]. Moreover, the theoretical framework of AQC has recently been recast into a variational setting [45]. The intuitive idea behind these approaches is that a full Hilbert space evolution may not be necessary to approximate the final ground state. In particular, when the final Hamiltonian is classical and with a glassy energy landscape, as in the typical framework of Eq. (11), obtaining a peaked wavefunction overlapping only with classical lowest-energy configurations may be both unfeasible and unnecessary. In fact, it would be enough to draw a sufficient number of samples from a wavefunction with a finite overlap with one or more of those classical solutions. In this heuristic setting, there is no evidence that a full Hilbert space evolution provides better results than a constrained evolution on a classically simulable variational manifold (see also Fig. 1 in Ref [45]). On the contrary, an evolution on a restricted manifold may even improve results in some cases [43].

In principle, variational simulations of QA can be achieved by solving real-time TDVP Eqs. (3) for a large annealing time τ , as theoretically characterized in Ref. [45] and numerically tested within a Matrix Product State [46] manifold. However, the numerical integration of TDVP equations is computationally demanding for ANNs and would require small time steps, making it extremely challenging for large system sizes and long

annealing times τ .

Variational Quantum Annealing (VQA) [3] is a different strategy that relies on iteratively *optimizing* the variational Ansatz to approximate the instantaneous adiabatic ground-state on the variational manifold. VQA leverages the variational principle to avoid solving real-time TDVP dynamics and may have the potential to mitigate the diverging runtime due to vanishing spectral gaps. Rather than physically evolving a quantum state, one uses a parametrized wavefunction and Variational Monte Carlo to approximately track the ground-state of $H(s)$ in Eq. (10) at a series of $N_A + 1$ discretized points $0 = s_0 < s_1 < \dots < s_{N_A} = 1$ along the annealing schedule. The algorithm realizes a discrete adiabatic path on the variational manifold by iteratively adjusting the Ansatz parameters to minimize the variational energy

$$E(s_j; \theta) = \frac{\langle \Psi(\theta) | H(s_j) | \Psi(\theta) \rangle}{\langle \Psi(\theta) | \Psi(\theta) \rangle}, \quad (12)$$

with $j = 0, \dots, N_A$. Optimizing the Ansatz parameters with standard SR is equivalent to performing repeated imaginary time evolutions for a set of parametric Hamiltonians $H(s_j)$.

VQA was originally presented in Ref. [3], where the authors devised both a variational version of classical Simulated Annealing and of Quantum Annealing. Both algorithms relied on Recurrent Neural Networks (RNNs) and were tested against standard techniques on several spin glass models. In this setting, VQA with RNNs appeared less effective than its classical counterpart, and VQA results were only shown for polynomial complexity problems, such as the random Ising chain and the 2D Edwards–Anderson model without longitudinal field.

Thanks to its flexibility, the VQA framework can naturally be viewed as an alternative approach to optimizing any NQS for ground-state search. This is useful when a direct optimization with SR on the target Hamiltonian H_T is expected to fail due to frustration (as in Eq. (11)), disorder, or other effects. In essence, the hardness of the ground-state problem is tuned from *easy* to *hard* while increasing the parameter s_j , and the NQS is warm-started with the result of the previous iteration (see Appendix A for details).

V. RESTRICTED AND DEEP BOLTZMANN MACHINES

Boltzmann machines (BMs) are undirected graphical models originally introduced as a general approach to learning probability distributions over binary variables [47]. BMs are energy-based models over a d -dimensional binary random vector $\mathbf{x} = \{-1, +1\}^d$ with a

quadratic energy function ²

$$E(\mathbf{x}; \boldsymbol{\theta}) = -\mathbf{x}^T W \mathbf{x} - \mathbf{b}^T \mathbf{x}, \quad (13)$$

where W and $\mathbf{b}$ are, respectively, the real-valued weight matrix and bias vector that we collectively denote with $\boldsymbol{\theta} = (W, \mathbf{b})$, and the corresponding probability distribution is given by ³

$$p(\mathbf{x}; \boldsymbol{\theta}) = Z^{-1} \exp\{\mathbf{x}^T W \mathbf{x} + \mathbf{b}^T \mathbf{x}\}, \quad (14)$$

where

$$Z = \sum_{\mathbf{x}'} \exp\{-E(\mathbf{x}'; \boldsymbol{\theta})\} \quad (15)$$

is the partition function. This class of models becomes particularly useful with the inclusion of hidden (unobserved) variables, which model higher-order interactions between the visible (observed) variables. With the inclusion of hidden variables, BMs become universal approximators of probability functions over discrete variables [48] and have been successfully used to solve a variety of machine learning problems [49–51]. For a more detailed discussion on standard BMs, we recommend Ref. [47].

Since BMs are notoriously hard to train and their probability distribution is hard to sample, a more practical and widely adopted alternative is the Restricted Boltzmann Machine (RBM), characterized by a single layer of visible variables and a single layer of hidden variables. No intra-layer interactions are included, resulting in a bipartite structure of the interaction graph where the only interactions occur between visible and hidden units. Formally, the visible layer consists of N_v visible variables that we collectively denote with $\mathbf{v}$, and the hidden layer consists of N_h hidden variables that we denote with $\mathbf{h}$. The energy function of the RBM is given by

$$E(\mathbf{v}, \mathbf{h}; \boldsymbol{\theta}) = -\mathbf{v}^T W \mathbf{h} - \mathbf{b}^T \mathbf{v} - \mathbf{c}^T \mathbf{h}, \quad (16)$$

with trainable parameters $\boldsymbol{\theta} = (W, \mathbf{b}, \mathbf{c})$.

The RBM architecture is generalized by introducing several layers of hidden variables, a model known as Deep Boltzmann Machine (DBM). In contrast to the RBM energy function, the hidden variables in a DBM interact with each other, albeit in a restricted manner, with only adjacent inter-layer connections and no intra-layer connections. By denoting the units in the l -th layer with $\mathbf{x}^{(l)}$, where layer 0 corresponds to the visible units and all hidden layers are indexed for $l > 0$, the energy function of a DBM with N_L layers reads

$$E(\mathbf{x}; \boldsymbol{\theta}) = - \sum_{l=0}^{N_L-2} \mathbf{x}^{(l)T} W^{(l)} \mathbf{x}^{(l+1)} - \sum_{l=0}^{N_L-1} \mathbf{b}^{(l)T} \mathbf{x}^{(l)}. \quad (17)$$

² To be consistent with the standard notation in quantum mechanics, we use spin variables with values ± 1 rather than conventional binary variables. The two conventions are equivalent up to an affine transformation.

³ To simplify the notation, we omit from now on the explicit dependence on $\boldsymbol{\theta}$ of the partition function Z .

Efficient training of the RBM and DBMs relies on block Gibbs sampling, enabled by the conditional independence of subsets of variables due to the bipartite graph structure. This is discussed in Appendix B.

VI. BOLTZMANN QUANTUM STATES

In this work, we introduce a novel framework to extend BMs to quantum variables, designed to represent quantum many-body states. We refer to this formulation as Boltzmann Quantum States (BQS). This class of states possesses a clear physical interpretation, as both hidden and visible variables are promoted to quantum spins over an extended Hilbert space.

Several prior extensions of Boltzmann Machines – particularly RBMs – to quantum settings have been proposed, as briefly reviewed in Appendix C. Our goal is to demonstrate that our approach has two key advantages compared to previous formulations. Firstly, BQS achieve a substantially higher expressive power, thanks to a natural extension from Restricted Boltzmann Quantum States (RBQS) to deep architectures (DBQS). Secondly, these new models naturally inherit block Gibbs sampling, allowing for efficient Monte Carlo estimation even in their deep formulations, with a speedup of orders of magnitude in runtime for the sampling part. In addition to these two key advantages, both RBQS and DBQS exhibit a further distinguishing feature: the calculation of the quantum geometric tensor and any local estimator is greatly simplified. In fact, while the former reduces to the computation of few-point correlators, the latter involves only simple matrix multiplications. This provides an additional computational speed-up for any involved optimization process.

A. General formulation

The definition of BQS is based on the idea of introducing unobserved (or hidden) *quantum* spins and using a Jastrow wavefunction to describe the state of the extended system. Physical observables act non trivially only on the Hilbert subspace related to physical (or visible) quantum spins. Formally, given a system of N_v visible spin-1/2 degrees of freedom with a Hamiltonian H_v , a Hilbert space $\mathcal{H}_v$, and computational basis $\{|\mathbf{v}\rangle\}$, we introduce another system of N_h hidden spin-1/2 degrees of freedom, Hilbert space $\mathcal{H}_h$, and computational basis $\{|\mathbf{h}\rangle\}$. We obtain an extended system with Hilbert space $\mathcal{H}_{vh} = \mathcal{H}_v \otimes \mathcal{H}_h$ and computational basis $\{|\mathbf{x}\rangle\} = \{|\mathbf{v}, \mathbf{h}\rangle\}$ on which we consider the Hamiltonian $H_{vh} = H_v \otimes \mathbb{I}_h$. As standard practice in NQS applications, we choose the Pauli-Z basis representation denoted as $|\mathbf{x}\rangle = |x_1, \dots, x_{N_u}\rangle$, with $N_u = N_v + N_h$ and $x_i \in \{-1; +1\}$ corresponding to spin down and spin up, respectively. Finally, a Jastrow variational wavefunction describes the state of the full system, effectively coupling

visible and hidden quantum spins:

$$\log \langle \mathbf{x} | \Psi(\boldsymbol{\theta}) \rangle = \sum_{ij} W_{ij} x_i x_j + \sum_i b_i x_i, \quad (18)$$

with $\boldsymbol{\theta} = (W, \mathbf{b})$, where $W \in \mathbb{C}^{N_u \times N_u}$ and $\mathbf{b} \in \mathbb{C}^{N_u}$ are complex variational parameters. We dub this family of Ansatz states as Boltzmann Quantum States (BQS). Note that this wavefunction is not normalized, as commonly done in NQS applications.

It can be easily shown that the probability function defined by a BQS is given by

$$\log[p(\mathbf{x}; \boldsymbol{\theta})] = \sum_{ij} 2 \operatorname{Re}[W_{ij}] x_i x_j + \sum_i 2 \operatorname{Re}[b_i] x_i, \quad (19)$$

which is the same probability function of a standard Boltzmann machine, with *real* parameters $\boldsymbol{\theta} = (2 \operatorname{Re}(W), 2 \operatorname{Re}(\mathbf{b}))$. This observation suggests that our Ansatz can be viewed as an over-parameterization of Boltzmann machines when addressing classical problems, specifically when the objective is to approximate a target probability distribution. However, when the objective is to approximate the ground-state or the dynamics of a quantum system as outlined in Section II, the full set of *complex* variational parameters $\boldsymbol{\theta} = (W, \mathbf{b})$ is used to optimize the Ansatz wavefunction in Eq. (18). Hence, BQS are a more general machine compared to standard BMs, as they can capture quantum correlations and entanglement. Compared to other architectures, such as Convolutional Neural Networks, BQS are particularly suitable for models defined on a graph without a clear 1D or 2D geometrical interpretation, or those that lack translational invariance, such as spin glasses or models with disorder.

By definition of the general BQS variational Ansatz in Eq. (18), visible and hidden quantum spins are entangled. As a consequence, it is not generally possible to marginalize the hidden spins to obtain a pure state for the visible ones. Nonetheless, the ground-state expectation value of an observable acting on the physical Hilbert space $\mathcal{H}_v$ can be computed on the ground-state of the extended system $\mathcal{H}_{vh}$ where the BQS is defined. This is discussed in Appendix D, where we also show that restricted and deep BQS architectures reduce to previous proposals of quantum BMs [2, 9] by projecting the hidden units on a specific product state.

B. Restricted and Deep Boltzmann Quantum States

Generic BQS suffer from the same intractability as Boltzmann machines. Nonetheless, it is straightforward to implement restrictions on the terms in the Jastrow wavefunction in Eq. (18). This allows to define restricted and deep Boltzmann quantum states, both of which enable efficient block Gibbs sampling and other useful properties.

As in the standard RBM, one can restrict the non-zero coefficients W_{ij} to connect only visible and hidden spins organized in two separate layers, leading to the variational wavefunction for the Restricted Boltzmann Quantum State (RBQS)

$$\log \langle \mathbf{v}, \mathbf{h} | \Psi(\boldsymbol{\theta}) \rangle = \mathbf{v}^T W \mathbf{h} + \mathbf{b}^T \mathbf{v} + \mathbf{c}^T \mathbf{h}. \quad (20)$$

The same notation as in Eq. (16) applies here, where $(\mathbf{v}, \mathbf{h})$ now label the computational basis states for the visible and hidden quantum spins. More generally, in close analogy to the case of DBMs, the spins can be arranged into layers $\mathbf{x}^{(l)}$, with the zero-th layer containing the visible spins, and all the successive layers including hidden quantum spins. By design, non-zero coefficients W_{ij} only connect spins in adjacent layers, and no intra-layer terms are present. Hence, the resulting Ansatz for the Deep Boltzmann Quantum State (DBQS) with N_L layers reads:

$$\log \langle \mathbf{x} | \Psi(\boldsymbol{\theta}) \rangle = \sum_{l=0}^{N_L-2} \mathbf{x}^{(l)T} W^{(l)} \mathbf{x}^{(l+1)} + \sum_{l=0}^{N_L-1} \mathbf{b}^{(l)T} \mathbf{x}^{(l)}. \quad (21)$$

Note that the DBQS Ansatz includes the RBQS as a special case. This is the class of variational wavefunction utilized in our work.

Due to Eq. (19), the probability distribution for the RBQS and the DBQS is the same as a canonical RBM and DBM, respectively. Hence, it factorizes conditionally on alternate layers, enabling the same efficient block Gibbs sampling routinely used by the machine learning community. A single Markov chain step updates all spins simultaneously, and numerical experiments show that autocorrelation times remain $\mathcal{O}(1)$ as the system size increases, in stark contrast to the superlinear scaling observed with local Metropolis moves. This is shown in Section D 4. Consequently, a speedup of at least an order of magnitude is achieved during both training and inference, allowing us to benchmark our methods for system sizes up to 700 spins.

C. Computational advantages

We now outline the other key conceptual innovations and computational advantages enabled by our formulation.

a. Analytical variational derivatives In a generic BQS formulation, the local derivatives O_k required to estimate the Fisher information matrix (see Eqs. (4)–(5)) reduce to simple products of spin variables and admit an analytical expression

$$\begin{aligned} \frac{\partial}{\partial W_{ij}} \log \langle \mathbf{x} | \Psi(\boldsymbol{\theta}) \rangle &= x_i x_j, \\ \frac{\partial}{\partial B_i} \log \langle \mathbf{x} | \Psi(\boldsymbol{\theta}) \rangle &= x_i. \end{aligned} \quad (22)$$

This represents a striking difference compared to NQS implementations based on feed-forward neural networks,

where computing gradients requires backpropagation, introducing additional computational complexity and memory overhead. Our approach avoids backpropagation, offering both conceptual simplicity and improved numerical stability. We note that the empirical Fisher information matrix can be efficiently constructed by computing connected correlation functions through sampling of the BQS wavefunction. For instance, the term of the S matrix associated with the parameters W_{ij} and W_{lm} reads

$$S_{ij,lm} = \langle x_i x_j x_l x_m \rangle - \langle x_i x_j \rangle \langle x_l x_m \rangle, \quad (23)$$

and similar equations can be easily derived for the other terms.

b. Evaluation of local operators Evaluating the expectation value of an operator O via Monte Carlo requires computing the local estimator defined in Eq. (7). In the context of spin models, operators can always be decomposed as a linear combination of Pauli strings, and the same holds for their local operators by linearity. For a system composed of N spins, a generic Pauli string operator is written as

$$P = \sigma_1^{\alpha_1} \otimes \sigma_2^{\alpha_2} \otimes \cdots \otimes \sigma_N^{\alpha_N}, \quad (24)$$

where each $\alpha_i \in \{0, x, y, z\}$, and $\sigma_i^{\alpha_i}$ denotes either a Pauli operator acting on spin i , or the identity operator $\sigma^0 = I$. The local estimator of a Pauli string has the general form of

$$P_{\text{loc}}(\mathbf{x}) = f(\mathbf{x}) \frac{\psi(\tilde{\mathbf{x}}; \boldsymbol{\theta})}{\psi(\mathbf{x}; \boldsymbol{\theta})} \quad (25)$$

where $f(\mathbf{x})$ is some function of the spin configuration $\mathbf{x}$ and $\tilde{\mathbf{x}}$ is the configuration connected to $\mathbf{x}$ by the Pauli string, i.e. $|\tilde{\mathbf{x}}\rangle \propto P |\mathbf{x}\rangle$.

In NQS applications based on a feed-forward ANN, computing P_{loc} requires one full forward pass for both configurations $\mathbf{x}$ and $\tilde{\mathbf{x}}$. For a BQS, the situation is radically different: inserting the explicit Ansatz of Eq. (18) into the ratio in Eq. (25) yields a simple exponential of differences of quadratic and linear forms

$$\frac{\Psi(\tilde{\mathbf{x}}; \boldsymbol{\theta})}{\Psi(\mathbf{x}; \boldsymbol{\theta})} = e^{\sum_{ij} W_{ij} (\tilde{x}_i \tilde{x}_j - x_i x_j) + \sum_i B_i (\tilde{x}_i - x_i)}. \quad (26)$$

In the specific case of a DBQS, when considering operators that act only on the visible spins, this expression simplifies even further

$$\frac{\Psi(\tilde{\mathbf{x}}; \boldsymbol{\theta})}{\Psi(\mathbf{x}; \boldsymbol{\theta})} = e^{\sum_i [(x_i^{(0)} - x_i^{(1)}) (b_i^{(0)} + \sum_j W_{ij}^{(0)} x_j^{(1)})]}, \quad (27)$$

and can be computed with simple tensor operations that can be easily vectorized to further reduce the time complexity of the computation. While the evaluation of local observables involving only the visible spins depends solely on the state of the visible units and the first hidden layer, the full set of hidden units remains necessary for sample generation via Gibbs sampling.

To better illustrate this idea, let us consider the case of the local energy for a spin chain of N sites and the Hamiltonian operator $H = \sum_{i=0}^{N-1} \sigma_i^x$. This also represents the only non-diagonal contribution for a standard QA Hamiltonian, as shown below in Eq. (10). In this case, the local energy of a given configuration, as defined in Eq. (5), reads

$$E_{\text{loc}}(\mathbf{x}; \boldsymbol{\theta}) = \sum_{i=0}^{N-1} \frac{\psi(T_i(\mathbf{x}); \boldsymbol{\theta})}{\psi(\mathbf{x}; \boldsymbol{\theta})} \quad (28)$$

where

$$T_i(\mathbf{x}) = (x_0, \dots, x_{i-1}, -x_i, x_{i+1}, \dots, x_{N-1}). \quad (29)$$

Hence, in a NQS implementation based on a feed-forward neural network, one needs to compute the output of the network for all possible single spin flips, requiring a total of $N+1$ evaluations of the network forward function. On the other hand, in the case of a DBQS, the local energy is simply given by

$$E_{\text{loc}}(\mathbf{x}; \boldsymbol{\theta}) = \sum_{i=0}^{N-1} e^{-2x_i^{(0)} (b_i^{(0)} + \sum_j W_{ij}^{(0)} x_j^{(1)})}. \quad (30)$$

Therefore, all the quantities needed to compute the local energy can be evaluated with a single matrix multiplication.

VII. NEURAL QUANTUM ANNEALING

Neural Quantum Annealing (NQA) is a novel algorithm that merges the high-level framework of VQA (see Section IV) with the expressive capabilities and sampling efficiency of DBQS (see Section VI), iteratively optimized with natural gradients and advanced stochastic optimizers. An illustrative representation of our algorithm is shown in Fig. 1, where the typical NQA optimization trajectory can be visualized on a reduced-dimensional space for a toy-size classical spin model. A preliminary version of our work using a standard cRBM wavefunction (see [2] and Appendix C) was presented in [44]. In the following, we describe our algorithm, highlighting its new features and the underlying theoretical motivations that distinguish our approach. A detailed pseudo-code and additional information on hyper-parameters are reported in Appendix E. Central to our method is the exploitation of the scalability and trainability of DBQS.

A. DBQS Ansatz

The primary challenge in sampling low-energy configurations of spin glasses lies in their highly rugged energy landscape, which results from energy frustration and competing interactions. This leads to a multitude of local minima that can trap heuristic optimization methods.

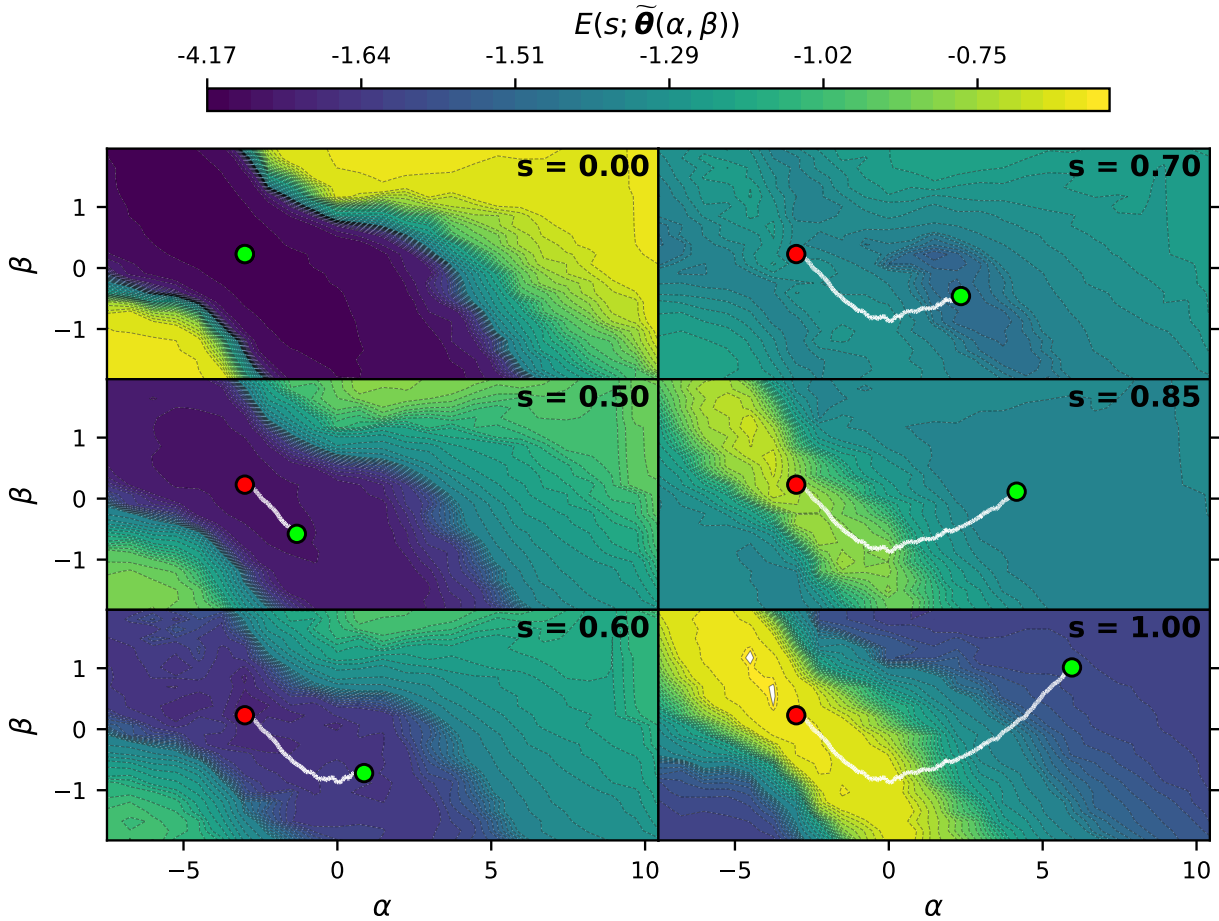


Figure 1. Visualization of the NQA trajectory for a 4-site Ising toy model with all-to-all random couplings. Using the technique proposed in Ref. [52], at each schedule point s_j , we record the instantaneous parameters and construct a real-valued data matrix by splitting each complex parameter into its real and imaginary parts and stacking them as separate features, i.e., $X_{j,(k,\mu)} \in \mathbb{R}$, with k spanning the θ components and $\mu \in \{\text{Re}, \text{Im}\}$. We then project the dynamics onto the affine PCA plane $\Sigma = \{\tilde{\theta}(\alpha, \beta) = \bar{\theta} + \alpha \mathbf{d}_1 + \beta \mathbf{d}_2\}$ spanned by the first two principal directions $\mathbf{d}_{1,2}$ of X [53]; (α, β) are the in-plane coordinates and $\bar{\theta}$ is the schedule mean. Each subplot shows the projected parameter trajectory (white line, with red and green dots marking the initial and current parameters) together with the instantaneous energy landscape on Σ at the corresponding s_j . Throughout the variational optimization, the Ansatz remains close to the minimum of the instantaneous energy, effectively guiding NQA to the global minimum of the target problem.

In particular, Markov Chain Monte Carlo (MCMC) algorithms relying on local update rules are effectively non-ergodic in the spin-glass phase and suffer from exponentially slow mixing due to large free-energy barriers [54], calling for more refined schemes [55]. Since most modern NQS architectures rely on Metropolis-based MCMC (local update rules), they are expected to perform poorly on spin glass sampling.

Here, we employ DBQS as a variational Ansatz. Unlike other NQS architectures, DBQS enable efficient block Gibbs sampling, which, thanks to its global update rule, is better suited for handling multimodal distributions in glassy energy landscapes. Moreover, DBQS can encode long-range, non-equivariant correlations on random graphs, providing both the expressivity and the sampling capabilities required for the problem. This makes our al-

gorithm viable for large-scale, complex spin glass problems where sampling is the main computational bottleneck. An alternative approach was proposed in Ref. [3], which employs recurrent neural network (RNN) wavefunctions that allow for autoregressive sampling. Although the RNN yields exact, uncorrelated samples, its capacity to represent the quantum many-body wavefunction during annealing is uncertain, especially for models on a random graph that lacks any 1D or 2D geometry, as the network must learn non-equivariant, long-range correlations. In practice, RNNs were found to be more conveniently optimized by thermal annealing [3]. This stands in contrast to several previous studies that have demonstrated the enhanced effectiveness of *simulated* quantum fluctuations over *simulated* thermal fluctuations [24, 56–59] in exploring rugged energy land-

scapes of classical frustrated models.

B. Persistent parallel Gibbs chains

Building on the sampling capability unlocked by the DBQS Ansatz, we run an ensemble of persistent Markov chains throughout the entire variational optimization. While a straightforward approach would be to generate and thermalize a new set of Markov chains at each annealing step $j = 0, \dots, N_A$ in Eq. (12), this would be a prohibitively slow procedure for high-dimensional glassy distributions.

Because successive Hamiltonians $H(s_j)$ and $H(s_{j+1})$ change only slightly, it is expected that thermalized chains at step j remain near equilibrium for the new distribution. Consequently, instead of fully re-thermalizing them, one can resume the chains from their last states and apply a few block Gibbs updates to adapt to the updated wavefunction parameters.

C. SR augmented with momentum

As anticipated in Section II, natural gradients can be combined with advanced stochastic optimizers, enhancing the standard SR algorithm for ground-state search. Rather than a straightforward optimization of the variational energy in Eq. (12), NQA combines natural gradients with a momentum term to perform the parameter updates. When using momentum [19–21], a velocity vector stores an exponentially decaying average of past gradients. Let us focus on a single annealing step j : for a fixed value of s_j , we perform a sequence of $n = 1, \dots, N_U$ optimization epochs to iteratively minimize the variational energy in Eq. (12). Concretely, at epoch n the updates are

$$\boldsymbol{\nu}_{n+1} = \mathbf{g}_n + \mu \boldsymbol{\nu}_n, \quad \boldsymbol{\theta}_{n+1} = \boldsymbol{\theta}_n - \eta \boldsymbol{\nu}_{n+1}, \quad (31)$$

where $\mathbf{g}_n$ is the natural gradient, $\boldsymbol{\nu}$ is the velocity vector, η is the learning rate, and $0 \leq \mu < 1$ is the momentum coefficient. The velocity term damps oscillations in directions where the gradient sign alternates while accelerating progress along directions with a persistent sign, often yielding faster and more stable convergence than standard gradient descent [60]. Crucially, both the optimized variational parameters and the final velocity obtained at annealing step j are carried over to warm start the DBQS Ansatz and the optimizer in the next annealing step $j+1$.

D. Relaxed adiabaticity on a manifold

While in the ideal limit – infinitely many annealing steps N_A , optimization epochs per annealing step N_U , and a sufficiently expressive Ansatz – VQA would recover the true adiabatic path in the whole Hilbert space,

it is clearly necessary to set a finite computational budget and select a NQS manifold with finite expressivity. The VQA framework is designed to remain close to the instantaneous variational ground-state, which is defined as the state that minimizes Eq. (12) as $H(s)$ slowly interpolates between H_0 and H_T . As recently pointed out in [45], it is essential to notice that the instantaneous *variational* ground-state does not need to coincide with the *exact* instantaneous ground-state of $H(s)$ at intermediate points s_j . Hence, the discretized manifold evolution does not, in general, closely approximate the ideal adiabatic dynamics in the full Hilbert space. Nonetheless, if the variational manifold is sufficiently expressive to capture both the initial ground-state of H_0 and the target ground-state of H_T , this manifold evolution may yield a valid solution – either the target quantum ground-state, or a wavefunction with finite overlap with classical energy minima of Eq. (11).

Additionally, even when approximating the adiabatic evolution on the manifold rather than in the whole Hilbert space, it is not necessary to find exact solutions of the intermediate ground-state problems. In fact, even if the Ansatz does not fully converge to the instantaneous variational ground-state at some intermediate step, this approximate solution will still serve as a valid warm start for the optimization problem in Eq. (12) at the next annealing step. Previous numerical errors can be compensated when re-optimizing the Ansatz, a feature that does not extend to real dynamics on a quantum annealer or real-time TDVP simulations. Guided by this observation, we recognize a trade-off in the choice of N_A and N_U , as the total computational cost scales linearly with their product. Rather than fully optimizing the variational parameters at each annealing step, it is often more efficient to increase the number of annealing steps N_A while performing only a few parameter updates N_U per step. In practice, even a single update per step ($N_U = 1$) can be sufficient to reach the final target ground state. To ensure that any residual excited-state components are suppressed at $j = N_A$, the variational parameters can be fine-tuned with a few additional optimization epochs as in Eq. (31).

E. Inclusion of a catalyst

The classical simulation of approximate adiabatic dynamics enables full flexibility in choosing Hamiltonian terms. We test the introduction of a catalyst H_C , which is an auxiliary Hamiltonian term vanishing at both $s = 0$ and $s = 1$:

$$H(s) = s H_T + A(1-s) H_0 + B s(1-s) H_C, \quad (32)$$

where H_0 is given by Eq. (9), and $A > 0$ and B are two tunable real hyper-parameters. By reshaping the instantaneous energy landscape, catalysts can convert a first-order phase transition encountered in real-time dynamics

– with an exponentially small spectral gap – into a higher-order transition with a polynomially closing gap [61, 62]. Similarly, it may help reduce diabatic errors during variational optimization in a VQA setting. In this work, we select

$$H_C = \sum_{j=1}^N \hat{\sigma}_j^y \quad (33)$$

as a catalyst, which is a non-stoquastic term that cannot be implemented on current quantum annealers [1].

VIII. NUMERICAL RESULTS

In this Section, we benchmark NQA on classical and quantum spin glass ground states, demonstrating its effectiveness for long-range models with frustration and NP-hard real-world problems. Further details on reproducibility are provided in the Code and Data Availability section.

A. Sherrington–Kirkpatrick model

We firstly address the Sherrington–Kirkpatrick (SK) model, a standard benchmark for testing algorithms designed to solve combinatorial optimization problems arising in science and industry, which often admit a reformulation as the energy minimization of similar spin models [28, 29]. A notable example is represented by resource allocation and scheduling problems, tackled in Section VIII B. The SK model is defined by N spin-1/2 degrees of freedom with all-to-all interactions whose coupling constants are sampled independently from a normal distribution. The Hamiltonian of the model reads

$$H = \frac{1}{\sqrt{N}} \sum_{i < j} J_{ij} \sigma_i^z \sigma_j^z, \text{ with } J_{ij} \sim \mathcal{N}(0, 1). \quad (34)$$

We prepared 10 realizations of the SK model with $N = 100$ spins and 10 realizations with $N = 200$ to benchmark our algorithm. Reference ground-state energies for these realizations were computed via 1000 runs of classical simulated annealing using OpenJij [63]. The spin-glass server [4], a standard tool for computing exact ground states of relatively small instances, was able to solve only 7 out of the 10 instances with $N = 100$ spins. In those cases, the results matched the lowest energies obtained with simulated annealing.

We are testing our methods on an NP-hard problem, where traditional exact solvers already struggle for $N \approx 100$, while $N = 200$ is considered an impractically large regime that requires approximate methods. Our novel DBQS architecture appears well-suited to address real-world NP-hard problems, which lack any geometrical structure and involve infinite-range interactions – features that are well captured by the SK model. In

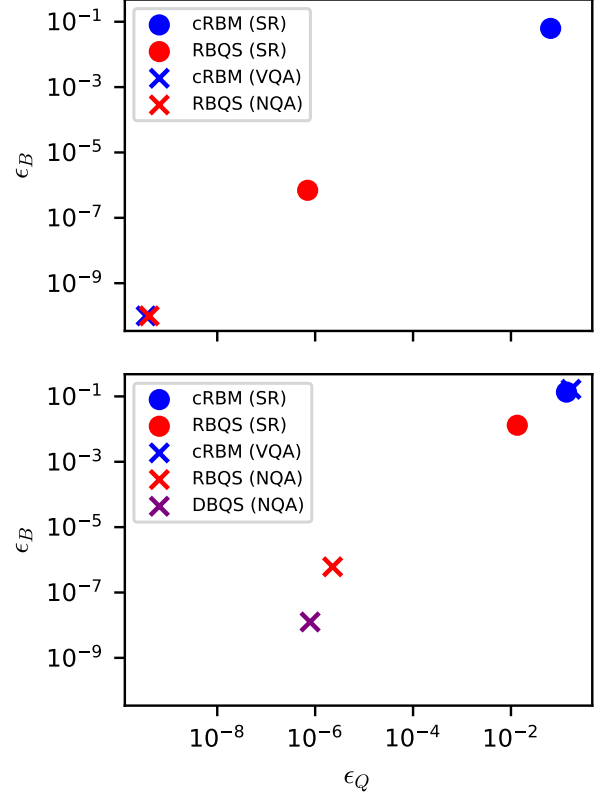


Figure 2. Comparison of the typical energy errors on the 10 realizations of the SK model with $N = 100$ (top panel) and $N = 200$ (bottom panel). Each point corresponds to a different combination of Ansatz and optimization protocol.

contrast, physical models are often characterized by finite dimensionality and symmetries, which can be leveraged to encode efficient inductive biases into NQS Ansatz design, as seen in the cases of CNNs and convolutional transformers for translationally invariant models, or 1D and 2D RNNs for low-dimensional systems.

For each run, we evaluate two relative energy error metrics to quantify solution quality: the final quantum state error $\varepsilon_Q \equiv (\langle H_T \rangle - E_0)/|E_0|$, and the energy error of the best classical state encountered during the simulation $\varepsilon_B \equiv (E_B - E_0)/|E_0|$, where E_0 is the exact ground state energy. For a classical spin glass, $\varepsilon_B = 0$ is a valid success criterion (a ground state was sampled during the simulation), even if the final wavefunction has not fully collapsed onto that or other ground states.

To aggregate results over multiple spin glass instances, we define the **typical (geometric mean)** errors as $\epsilon_Q = [\varepsilon_Q]_{\text{typ}}$ and $\epsilon_B = [\varepsilon_B]_{\text{typ}}$, where $[x]_{\text{typ}} \equiv \exp(\langle \log x \rangle)$ is the exponential of the average log-error (with a 10^{-10} floor for numerical stability) and the average $\langle \cdot \rangle$ is taken over instances.

We first test the performance of a cRBM with real parameters (see [2] and Appendix C) optimized via SR

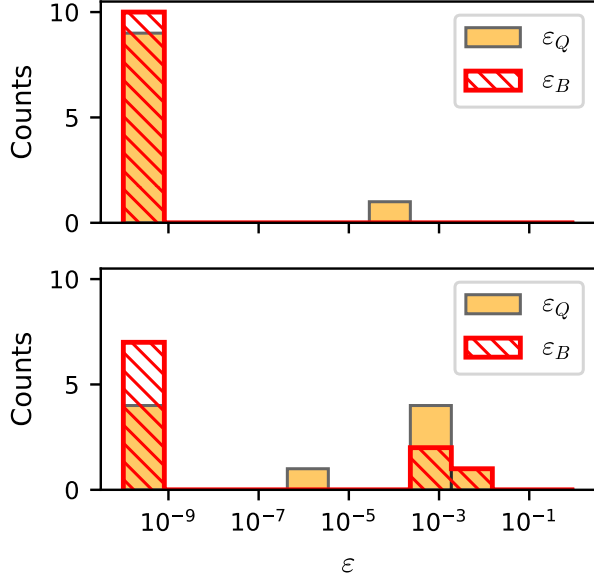


Figure 3. Residual energies histograms on the SK instances for $N = 100$ (top panel) and $N = 200$ (bottom panel) with the best performing settings. Our set-up manages to exactly solve all the $N = 100$ spins instances and 7 out of 10 of the $N = 200$ spins instances.

(natural gradients with momentum) or VQA, see Appendix A for details on the algorithms. We compare it against RBQS with real parameters, optimized with SR (natural gradients with momentum), or in the framework of NQA (Appendix E). The use of real-valued wavefunctions is justified since the QA Hamiltonian in Eq. (32) is stoquastic for $B = 0$, and for a standard driver and target Hamiltonian in Eq. (9) and Eq. (11), respectively; hence, its instantaneous ground state can be described by a (positive) real-valued wavefunction. Sampling is performed using block Gibbs updates for RBQS or single-spin-flip Metropolis-Hastings chains for cRBM, in both cases kept persistent across annealing steps. To ensure a fair comparison, we conduct a brief hyperparameter optimization (HPO) using Optuna [64], allocating the same number of trials and wall-clock budget per trial to each combination of Ansatz and optimization algorithm. Details about HPO⁴ are reported in Appendix E.

By construction, $0 \leq \epsilon_B \leq \epsilon_Q$, and $\epsilon_B = \epsilon_Q = 0$ signals that the algorithm converges to a state overlapping only with ground-state configuration(s) of a problem instance. Figure 2 compares the performance of the different set-ups according to the typical energy errors introduced above.

The SR algorithm performs poorly on this benchmark, regardless of whether momentum is included; the cRBM ansatz fails to find the ground state for any instance, as highlighted by large values of both typical metrics. The RBQS Ansatz solves exactly (i.e. $\epsilon_B = 0$) only half of the instances, as shown in ... cite the appendix with histograms here. The results are drastically improved when using VQA for the cRBM and NQA for the RBQS: in both cases, all instances with $N = 100$ spins are solved exactly, including those unsolved by the spin-glass server. These results demonstrate that the guided annealing strategy is significantly more effective in navigating the rugged energy landscape of spin glasses than direct SR optimization. We remark that the number of HPO trials and the wall-clock budget per trial are set equal for a fair algorithmic comparison. **TODO**.

When scaling up to $N = 200$ spins, the problem difficulty increases drastically. Only the combination of VQA and RBQS manages to solve half of the instances, with typical residual energies of $\epsilon_B = 4.953 \times 10^{-07}$ and $\epsilon_Q = 1.788 \times 10^{-06}$, while the other combinations fail to solve any of the instances. Performance on the $N = 200$ instances, as measured by ϵ_B , can be improved with the inclusion of a catalyst, and complex-valued parameters to describe the non-stoquastic nature of the instantaneous ground state. This catalyzed version of VQA yields typical residual energies of $\epsilon_B = 6.359 \times 10^{-08}$ and $\epsilon_Q = 2.954 \times 10^{-06}$. The final and most significant gains come from increasing the expressiveness of the variational state itself. By employing NQA, a DBQS Ansatz with a second hidden layer within the catalyzed VQA framework, we achieved the highest success rate and accuracy. In this configuration, our algorithm attained the exact ground state in 7 out of 10 instances at $N = 200$, yielding typical residual energies of $\epsilon_B = 1.256 \times 10^{-08}$ and $\epsilon_Q = 7.883 \times 10^{-07}$.

Figure 3 presents the histograms of the residual energy errors on the benchmark instances for our best performing set-ups.

In summary, our Neural Quantum Annealing strategy consistently outperforms both conventional variational optimization and prior quantum-inspired methods on the classical SK model. We observe that (i) guiding the variational optimization via simulated quantum annealing yields systematically lower energy errors than stochastic reconfiguration even when augmented with the inclusion of more advanced optimizers, (ii) at $N = 100$ enabled all test instances to be solved exactly, and (iii) at $N = 200$, the combination of a catalyzed VQA protocol with a DBQS Ansatz delivered the best overall results, markedly surpassing the next-best approach (RBQS with VQA) in both success probability and residual energy. These results highlight the scalability and raw performance benefits of integrating quantum-annealing-inspired dynamics with expressive deep network wavefunctions.

⁴ Note that standard SR optimization is included within our HPO as the limit of zero momentum, which is consistently outperformed by non-zero momentum values.

B. Large-scale JSSP instance

Missing: explicit JSSP mapping to Ising and variable pruning. Moreover, without the energy shift the gs has -something energy, so clearly some contribution is negative. To benchmark our set-up on large-scale classical Ising spin glasses we turn to the Ising formulation of the Job Shop Scheduling Problem (JSSP). Instances of the JSSP can be formulated as ground-state problems for Ising spin glasses similar to the SK model, but with a critical difference: the ground-state energy (not the configuration, though) is known *a priori* by construction when one encodes the objective as counting constraint violations. Such problems provide an ideal benchmark at larger sizes: we can assert convergence to the true ground state (zero violations) without relying on other approximate solvers. In our formulation, the energy of a spin configuration equals the number of scheduling constraints that are violated in the corresponding schedule. The QUBO encodes (i) machine exclusivity, (ii) precedence constraints within each job, and (iii) execution of all jobs, with penalty weights chosen such that any feasible schedule has energy 0 and any infeasible schedule has a positive integer energy proportional to the number of violated constraints.

We evaluate NQA on a set of 10 standard 5×5 JSSP instances. For each instance, we run the same evaluation protocol used in the SK benchmarks, tracking both the final variational energy and the best classical configuration encountered, and we report success when a zero-violation schedule (ground state) is sampled at any point during the run. Across this benchmark, NQA achieves high success rates and very low residual energies when it fails to reach zero violations, indicating near-optimal schedules.

Beyond this benchmark set, we also scaled to substantially larger JSSP encodings, reaching problem sizes mapped to up to $N = 725$ spins. These larger runs demonstrate that DBQS sampling and the NQA optimization protocol remain stable and effective well into the hundreds-of-spins regime on structured QUBOs.

To complement the aggregate metrics, we include a histogram summarizing the number of instances solved exactly and the residual violations for the remaining cases (Figure ??). We also provide a representative convergence trace for a single instance (Figure 4), showing the decay of the best violation count over the annealing schedule.

C. Transverse field Sherrington-Kirkpatrick model

To benchmark our set-up on quantum spin glasses, we consider the Sherrington-Kirkpatrick model in a transverse field, i. e.:

$$H = \sum_{i < j} J_{ij} \sigma_i^z \sigma_j^z + g \sum_i \sigma_i^x, \quad (35)$$

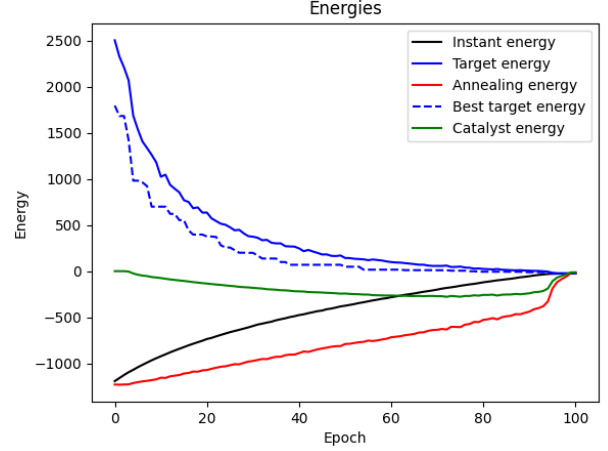


Figure 4. Representative JSSP instance: convergence of the best violation count observed during NQA as a function of the annealing schedule step.

with

$$J_{ij} \sim \mathcal{N}(0, 1/N). \quad (36)$$

Specifically, we analyze the same disorder realizations of the SK model with $N = 100$ spins used in the previous benchmark. We choose transverse field $g=0.1$ to make sure we are deep in the spin glass phase. In the absence of an external reference for validation, we assess convergence to the ground state via internal consistency checks. To characterize the TFSK ground state, a straightforward variational estimate using the classical ground state yields an upper bound that lies at the classical SK ground-state energy. Consistent with practice in machine-learning and Monte Carlo, a single optimization run is insufficient, and convergence must be examined with respect to the tunable hyperparameters. In this setting, HPO serves a dual role: it identifies near-optimal hyperparameters and simultaneously provides the data needed to substantiate convergence to the ground state. In our case, we use CMA-ES and TPE for hyperparameter optimization (100 runs of CMA-ES followed by 50 of TPE).

Across all 10 disorder realizations we find final energies below the classical counterpart, and the optimization exhibits consistent convergence toward an eigenstate compatible with the ground state, as indicated by the concurrent decrease of energy and energy variance. Eigenstates have zero energy variance, and for a small perturbation of an eigenstate, the energy and energy variance are tied by a linear relationship. The energy–variance scatter plot in Fig. 5 (left panel) reveals multiple, well-separated energy bands, each corresponding to a distinct eigenstate. Zero-variance extrapolation performed within each band yields estimates of the associated low-lying eigenvalues, thereby reconstructing a portion of the low-energy spectrum. Fig. 5 showcases the results of HPO for the first of

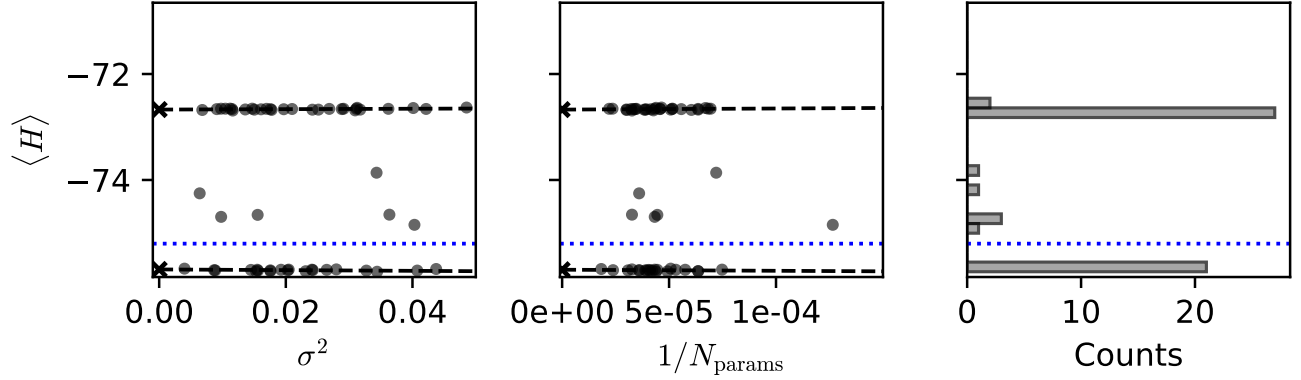


Figure 5. Final energy vs final energy variance and inverse number of network parameters for instance 0 found optimizing DBQSs with two hidden layers and variable number of units via NQA. The blue dotted line represents the classical energy ($g=0$) for that instance. The transverse field for this optimization is $g=0.1$. The center panel shows final energy as a function of final energy variance, while the right panel shows final energy versus the inverse number of parameters N_{params} . The 'X' marks indicate extrapolated energies: zero-variance extrapolation in the left panel and infinite-parameter extrapolation in the right panel. Only points with variance below 0.05 are considered to filter states that can not be considered perturbatively close to an eigen state. Right panel shows the histogram of the energies found. Monte Carlo error bars, being small enough, are omitted as they would not be visible. Other than finding the ground state, in this case we can clearly see the presence of another low energy eigenstate. Also fit is weighted, with $w=1/\sqrt{\text{variance}+1e-3}$ (i.e.: weighted accordingly to the MC error since all the trials have the same number of parameters, plus small positive constant for numerical stability). **Only bands with more than 5 points are considered.**

our benchmark instances (results for the other instances are reported in G).

Zero-variance extrapolation details. **should explain how we identify the bands. Someone who reads this would not understand how to reproduce this analysis.** For each instance, we partition the energy axis into narrow splices. A band is identified whenever multiple optimized energy points fall within the same splice, as illustrated in Fig. 5, where clusters of points reveal well-defined bands. **reason of choosing 5 is ambiguous?** Whenever five or more points appear in a splice, we perform a variance extrapolation: the data are ordered by variance, and a linear fit $E(\text{Var}) = E_0 + a \text{Var}$, is applied to the lowest-variance subset. The intercept E_0 is taken as the zero-variance estimate, which remains stable under modest variations of the slice size or cutoff.

Naturally, the same analysis can be carried out using the inverse of the number of parameters instead of the energy variance, where the same band structure appears. We also show the infinite parameter extrapolated energy for the distinct energy bands in Fig. 5.

IX. DISCUSSION AND OUTLOOK

Summarize again the key results. Our results demonstrate that combining DBQS with a variational annealing strategy yields state-of-the-art performance on challenging spin-glass benchmarks. On classical SK instances up to $N = 200$, NQA provide the best overall perfor-

mance. On quantum TF-SK instances, energy-variance analyses across HPO trials indicate convergence to the ground state and even allow low-lying spectra to be reconstructed via zero-variance extrapolation.

Outlook. Several avenues appear promising: (a) Adaptive NQA schedules guided by variational Fubini-Study distances between successive states; (b) Parallel-tempering variants that swap persistent chains across schedule points to mitigate mode collapse; (c) Convolutional or attention-based DBQS to inject inductive biases for structured problems; (d) Extensions beyond quadratic Ising-like models; (e) Systematically studying the phase transition of the TFSK model.

X. CODE AND DATA AVAILABILITY

All code to reproduce SK, TF-SK, and JSSP experiments, along with scripts to generate figures and tables, will be released in a public repository at: **<to be added>** (commit/tag: **<to be added>**). Hyperparameter configurations and random seeds used for the main figures will be provided under **experiments/**. Aggregate results and raw logs will be archived on an open data repository (e.g., Zenodo) at: **<to be added>**.

XI. COMPUTATIONAL RESOURCES

Cite ALCC and LiCCA. Experiments were run on a Linux cluster with CUDA-enabled GPUs and multi-core CPUs. Typical HPOs with NQA completed within a day on a single A100 GPU, with TF-SK HPO runs requiring slightly larger trial budgets to collect the statistics required to assert convergence to the ground-state.

XII. RINGRAZIAMENTI

Thanks to Bojan, Marcus K, Arnab Das .

Appendix A: Algorithmic details on NQS optimization for ground-state search

1. Natural gradients and advanced optimizers

In Section II, we introduced the real and imaginary-time TDVP and defined the Fisher information matrix. A noteworthy observation is that imaginary-time TDVP is formally equivalent to the optimization of the cost function in Eq. (8) via natural gradient descent, a seminal algorithm known as Stochastic Reconfiguration (SR) [17], recently improved by min-SR [7]. This equivalence follows from the definition of natural gradient, obtained by applying the inverse Fisher information matrix to the standard gradient:

$$\mathbf{g}_\theta = S^{-1}(\theta) \nabla_\theta. \quad (\text{A1})$$

Indeed, a first-order numerical integration of the imaginary-time TDVP equations leads to an iterative parameter update

$$\theta(\tau + \delta\tau) = \theta(\tau) - \delta\tau S^{-1}(\theta(\tau)) F(\theta(\tau)), \quad (\text{A2})$$

which corresponds to a Natural Gradient Descent step with step size $\delta\tau$ since

$$\mathbf{g}_\theta E(\theta) = S^{-1}(\theta) \nabla_\theta E(\theta) = S^{-1}(\theta) F(\theta). \quad (\text{A3})$$

As briefly mentioned in Section II, natural gradients can be combined with advanced stochastic optimizers developed for deep learning. While this strategy has not been explored in the NQS literature to our knowledge, it has been studied in the broader machine learning community. For example, works such as [65, 66] precondition the stochastic gradient with an (approximate) inverse Fisher information matrix and apply momentum before updating parameters. Ref. [67] further proposes a unified framework for accelerated natural gradient descent.

In practice, an optimizer maintains an auxiliary state α , and given the (natural) gradient, returns both the parameter update and the new auxiliary state. As stated in

Section VII, in this work, we combine SR with momentum, where the auxiliary state is a velocity vector storing an exponentially decaying average of past natural gradients.

Algorithm 1 presents a high-level overview of NQS optimization for ground-state search, potentially combining natural gradients with an advanced optimizer. As remarked in Section II, we collectively refer to all these variants as SR in the remainder of the text.

Algorithm 1 SR for ground-state search

Require: Number of training epochs N_ϵ ; Optimizer $\text{opt}(\mathbf{g}, \alpha)$; Variational Ansatz $\psi(\mathbf{x}; \theta)$.

Ensure: Optimized parameters θ

- 1: Initialize parameters $\theta \leftarrow \theta_0$
 - 2: Initialize auxiliary state of the optimizer $\alpha \leftarrow \alpha_0$
 - 3: **for** $n = 0$ **to** $N_\epsilon - 1$ **do**
 - 4: Sample configurations $\{\mathbf{x}\}$ from $p(\mathbf{x}) \propto |\psi(\mathbf{x}; \theta)|^2$
 - 5: Compute the natural gradient $\mathbf{g} \leftarrow \mathbf{g}_\theta E(\theta)$
 - 6: Compute the updates for the parameters and the new optimizer state $(\delta\theta, \alpha) \leftarrow \text{opt}(\mathbf{g}, \alpha)$
 - 7: Update the variational parameters $\theta \leftarrow \theta + \delta\theta$
 - 8: **end for**
 - 9: **return** θ
-

2. Variational Quantum Annealing pseudo-code

As explained in Section IV, an alternative approach to ground-state optimization with NQS relies on interpolating from an *easy* to a *hard* regime by tuning a parameter in the Hamiltonian. The resulting high-level implementation of VQA for optimizing a variational Ansatz (typically an NQS) is illustrated in Algorithm 2. We note that, in Ref. [3], the parametric Hamiltonian is defined slightly differently compared to Eq. (10), precisely as $H(s) = H_T + (1 - s)H_0$, with $H_0 = -B^x \sum_i \sigma_i^x$ and $B^x \gg 1$. Although Ref. [3] uses Adam and simple gradients to train RNN wavefunctions, it is possible to extend this approach to other NQS architectures and to natural gradients.

Algorithm 2 VQA for ground-state search

Require: Number of warm-up steps N_W ; Number of annealing steps N_A ; Number of updates per annealing step N_U ; Optimizer $\text{opt}(\mathbf{g}, \alpha)$ (Adam and simple gradients in [3]); Variational Ansatz $\psi(\mathbf{x}; \boldsymbol{\theta})$ (RNN in [3]).

Ensure: Optimized parameters $\boldsymbol{\theta}$

```

1: Initialize parameters  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}_0$ 
2: Initialize auxiliary state of the optimizer  $\alpha \leftarrow \alpha_0$ 
3:  $H(s) \leftarrow H_0$ 
4: for  $n = 0$  to  $N_W - 1$  do
5:   Sample configurations  $\{\mathbf{x}\}$  from  $p(\mathbf{x}) \propto |\psi(\mathbf{x}; \boldsymbol{\theta})|^2$ 
6:   Compute (natural) gradients of the variational energy
    $\mathbf{g} \leftarrow \nabla_{\boldsymbol{\theta}} E(s; \boldsymbol{\theta})$ 
7:   Compute the updates for the parameters and the new
   optimizer state  $(\delta\boldsymbol{\theta}, \alpha) \leftarrow \text{opt}(\mathbf{g}, \alpha)$ 
8:   Update the variational parameters  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \delta\boldsymbol{\theta}$ 
9: end for
10: for  $j = 1$  to  $N_A$  do
11:    $s \leftarrow j/N_A$ 
12:    $H(s) \leftarrow sH_T + (1-s)H_0$ 
13:   for  $n = 0$  to  $N_U - 1$  do
14:     Sample configurations  $\{\mathbf{x}\}$  from  $p(\mathbf{x}) \propto |\psi(\mathbf{x}; \boldsymbol{\theta})|^2$ 
15:     Compute (natural) gradients of the variational en-
     ergy  $\mathbf{g} \leftarrow \nabla_{\boldsymbol{\theta}} E(s; \boldsymbol{\theta})$ 
16:     Compute the updates for the parameters and the
     new optimizer state  $(\delta\boldsymbol{\theta}, \alpha) \leftarrow \text{opt}(\mathbf{g}, \alpha)$ 
17:     Update the variational parameters  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \delta\boldsymbol{\theta}$ 
18:   end for
19: end for
20: return  $\boldsymbol{\theta}$ 

```

Appendix B: Standard analytical results on Boltzmann Machines

In Section V, we introduced the RBM as the simplest efficiently trainable version of generic BMs. Starting from Eq. (16), the joint probability function over visible and hidden variables reads

$$p(\mathbf{v}, \mathbf{h}; \boldsymbol{\theta}) = Z^{-1} \exp\{\mathbf{v}^T W \mathbf{h} + \mathbf{b}^T \mathbf{v} + \mathbf{c}^T \mathbf{h}\}, \quad (\text{B1})$$

where

$$Z = \sum_{\mathbf{v}', \mathbf{h}'} \exp\{-E(\mathbf{v}', \mathbf{h}'; \boldsymbol{\theta})\}. \quad (\text{B2})$$

This partition function Z has been proven to be intractable [68], implying that the explicit computation of a normalized marginal probability distribution for the visible units is also intractable. Nonetheless, it is possible

to exactly trace out the hidden variables to obtain the unnormalized marginal distribution

$$\begin{aligned} p_{\text{RBM}}(\mathbf{v}; \boldsymbol{\theta}) &= \sum_{\mathbf{h}} p(\mathbf{v}, \mathbf{h}; \boldsymbol{\theta}) \\ &= Z^{-1} \exp\{\mathbf{b}^T \mathbf{v}\} \prod_{j=0}^{N_h} \cosh \left\{ c_j + \sum_{i=0}^{N_v} v_i W_{ij} \right\}. \end{aligned} \quad (\text{B3})$$

The inability to compute the exact partition function does not constitute a serious limitation, as multiple methods have been devised to approximate it [47], and its explicit computation is often unnecessary in practice: Monte Carlo methods circumvent the problem of computing the partition function by computing unweighted averages over the samples drawn from the model.

The key property of RBMs that enables efficient sampling and training stems from the bipartite structure of their interaction graph: conditioned on the hidden units, the visible units are mutually independent, and vice versa. This allows us to explicitly compute the normalized conditional probabilities

$$\begin{aligned} p(\mathbf{v}|\mathbf{h}; \boldsymbol{\theta}) &= \frac{p(\mathbf{v}, \mathbf{h}; \boldsymbol{\theta})}{p(\mathbf{h}; \boldsymbol{\theta})} \\ &= \frac{1}{Z} \frac{\exp\{\mathbf{c}^T \mathbf{h}\}}{p(\mathbf{h}; \boldsymbol{\theta})} \exp\{\mathbf{b}^T \mathbf{v} + \mathbf{v}^T W \mathbf{h}\} \\ &= \frac{1}{Z'} \exp\{\mathbf{b}^T \mathbf{v} + \mathbf{v}^T W \mathbf{h}\} \\ &= \frac{1}{Z'} \prod_{i=1}^{N_v} \exp \left\{ \left(b_i + \sum_{j=1}^{N_h} W_{ij} h_j \right) v_i \right\} \\ &\equiv \frac{1}{Z'} \prod_{i=1}^{N_v} \tilde{p}(v_i | \mathbf{h}; \boldsymbol{\theta}). \end{aligned} \quad (\text{B4})$$

Here, the hidden variables are treated as constants, as we are conditioning on their values. Hence, all terms not explicitly depending on visible units are included in a new constant Z' . The conditional probability distribution is factorized over the visible units, and the single terms can be easily normalized since computing the partition function in this case involves a sum over two configurations only. The normalized conditional probabilities read:

$$\begin{aligned}
p(v_i|\mathbf{h};\boldsymbol{\theta}) &= \frac{\tilde{p}(v_i|\mathbf{h};\boldsymbol{\theta})}{\tilde{p}(v_i=+1|\mathbf{h};\boldsymbol{\theta}) + \tilde{p}(v_i=-1|\mathbf{h};\boldsymbol{\theta})} \\
&= \frac{\exp\left\{\left(b_i + \sum_{j=1}^{N_h} W_{ij}h_j\right)v_i\right\}}{\exp\left\{\left(b_i + \sum_{j=1}^{N_h} W_{ij}h_j\right)\right\} + \exp\left\{-\left(b_i + \sum_{j=1}^{N_h} W_{ij}h_j\right)\right\}} \\
&= \exp\left\{\left(b_i + \sum_{j=1}^{N_h} W_{ij}h_j\right)(v_i - 1)\right\} \sigma\left[2\left(b_i + \sum_{j=1}^{N_h} W_{ij}h_j\right)\right],
\end{aligned} \tag{B5}$$

where $\sigma(x)$ denotes the logistic sigmoid function. The normalized probability for the i -th visible unit to have value $+1$, conditioned on all hidden variables, simplifies to

$$p(v_i=+1|\mathbf{h};\boldsymbol{\theta}) = \sigma\left[2\left(b_i + \sum_{j=1}^{N_h} W_{ij}h_j\right)\right], \tag{B6}$$

and a similar expression holds for the value -1 . Since the difference between visible and hidden units is only formal, the conditional probability for the hidden units given the visible ones can be computed in the same way:

$$p(h_j=+1|\mathbf{v};\boldsymbol{\theta}) = \sigma\left[2\left(c_j + \sum_{i=1}^{N_v} W_{ij}v_i\right)\right]. \tag{B7}$$

The mutual independence of visible units conditioned on hidden units, and vice versa, enables efficient block Gibbs sampling of the joint probability function $p(\mathbf{v}, \mathbf{h}; \boldsymbol{\theta})$. By starting from a random configuration, a Markov chain can be constructed with the update rule that, at each iteration, all visible units are simultaneously sampled conditioned on the hidden units, and then all hidden units are sampled conditioned on the visible units. Moreover, these samples can be drawn via exact sampling of the conditional probability in Eqs. (B6) and (B7). Such an alternating conditional sampling efficiently generates samples from the joint probability distribution $p(\mathbf{v}, \mathbf{h}; \boldsymbol{\theta})$ [47, 69], which are necessary both during training and inference.

During training, the RBM parameters $\boldsymbol{\theta}$ are typically optimized by minimizing the negative log-likelihood using variants of gradient descent. The gradient of such objective function can be expressed in terms of expectation values and two-point correlations between hidden and visible units [70], which are efficiently estimated via Gibbs sampling. At the inference stage, the trained model would be used to compute the expectation values of functions of the visible units $f(\mathbf{v})$ over the marginal probability distribution $p_{\text{RBM}}(\mathbf{v}; \boldsymbol{\theta})$ in Eq. (B3), which cannot be explicitly normalized. This expectation value, denoted as $\langle \cdots \rangle_{\text{RBM}}$, is equal to the expectation value of

the same function over the joint probability $p(\mathbf{v}, \mathbf{h}; \boldsymbol{\theta})$:

$$\begin{aligned}
\langle f(\mathbf{v}) \rangle_{\text{RBM}} &= \frac{\sum_{\mathbf{v}} p(\mathbf{v}) f(\mathbf{v})}{\sum_{\mathbf{v}} p(\mathbf{v})} \\
&= \frac{\sum_{\mathbf{v}, \mathbf{h}} p(\mathbf{v}, \mathbf{h}) f(\mathbf{v})}{\sum_{\mathbf{v}, \mathbf{h}} p(\mathbf{v}, \mathbf{h})} = \langle f(\mathbf{v}) \rangle_{p(\mathbf{v}, \mathbf{h})}.
\end{aligned} \tag{B8}$$

This allows us to efficiently estimate it by performing block Gibbs sampling and using the normalized conditional probability functions in Eqs. (B6) and (B7).

The more general case of DBMs in Eq. (17) is similarly treated. The joint probability function is given by

$$p(\mathbf{x}; \boldsymbol{\theta}) = \frac{1}{Z} e^{\sum_{l=0}^{N_L-1} \mathbf{x}^{(l)T} \mathbf{W}^{(l)} \mathbf{x}^{(l+1)} + \sum_{l=0}^{N_L} \mathbf{b}^{(l)T} \mathbf{x}^{(l)}}. \tag{B9}$$

where

$$Z = \sum_{\mathbf{x}'} \exp\{-E(\mathbf{x}'; \boldsymbol{\theta})\}. \tag{B10}$$

The computation of the marginal probability function for the visible units, as done for the RBM in Eq. (B3), is now intractable even in its unnormalized version. Nonetheless, similarly to the case of RBMs, the variables can be rearranged into a bipartite graph, with odd layers on one side and even layers on the other. This immediately implies that when we condition on the units in the even layers, the variables in the odd layers become independent, and vice versa. More precisely, it is enough to condition on the adjacent layers, and the explicit computation of the normalized conditional probability leads to

$$\begin{aligned}
p(x_i^{(l)} = +1 | \mathbf{x}^{(l-1)}, \mathbf{x}^{(l+1)}; \boldsymbol{\theta}) &= \\
&\sigma\left[2\left(\sum_j W_{ji}^{(l-1)} x_j^{(l-1)} + \sum_j W_{ij}^{(l)} x_j^{(l+1)} + b_i^{(l)}\right)\right],
\end{aligned} \tag{B11}$$

with slightly different expressions for the first and last layers. Analogously to the RBM case, this conditional independence enables efficient block Gibbs sampling by updating all units in only two iterations. Firstly, one can update all the units in the even layers (including the $l=0$ visible layer) as a block, conditioned on the units in the odd layers, and, secondly, all the hidden units in the odd layers conditioned on the units in the even ones. The samples can be drawn via exact sampling of the conditional probability in Eq. (B11).

Appendix C: Previous quantum extensions of Boltzmann Machines

Several extensions of BMs and specifically RBMs to quantum spin systems have been previously proposed. For quantum systems, the fundamental object is not the probability function, but rather the wavefunction, which is a complex-valued function representing the probability amplitudes.

1. pmRBM and cRBM

A conceptually straightforward extension proposed in Ref. [71] and known as pmRBM uses two independent RBMs to model the phase and the modulus of the wavefunction. The wavefunction is expressed in the spin eigenbasis, which is commonly associated with the eigenstates of the Pauli-Z operators:

$$\psi(\mathbf{x}; \boldsymbol{\theta}) = \sqrt{p_{\text{RBM}}^m(\mathbf{x}; \boldsymbol{\theta}_m)} \exp\{i p_{\text{RBM}}^p(\mathbf{x}; \boldsymbol{\theta}_p)\}, \quad (\text{C1})$$

where $p_{\text{RBM}}(\mathbf{x}; \boldsymbol{\theta})$ denotes the RBM unnormalized marginal probability distribution (see Appendix B). Here, $\boldsymbol{\theta} = (\boldsymbol{\theta}_m, \boldsymbol{\theta}_p)$ are real parameters, and $\mathbf{x}$ is a collective label for a spin eigenstate in the Pauli-Z eigenbasis. Current literature suggests that the artificial separation of modulus and phase information is suboptimal, making the learning problem more challenging, regardless of the specific ANN model used. Almost all recent NQS applications [14] utilize ANNs parameterized by complex variational parameters to output a complex number representing the wavefunction amplitude. This approach was proposed in the context of RBMs in Ref. [2], which is based on the idea of promoting the probability distribution defined by an RBM to quantum amplitudes by introducing complex parameters. The resulting Ansatz, commonly referred to as cRBM, reads

$$\psi(\mathbf{x}; \boldsymbol{\theta}) = e^{\sum_{i=0}^{N_v} b_i x_i} \prod_{j=0}^{N_h} \cosh \left\{ c_j + \sum_{i=0}^{N_v} x_i W_{ij} \right\}, \quad (\text{C2})$$

which is formally equivalent to the unnormalized marginal probability distribution of an RBM (see Appendix B), with the only difference that $\boldsymbol{\theta} = (W, \mathbf{b}, \mathbf{c})$ are now complex parameters.

In combination with the standard SR algorithm, this Ansatz has been successfully used to find a variety of quantum ground-states with remarkable precision [2, 72–74] and in the context of NQA in Ref. [44]. However, the possibility of efficient block Gibbs sampling is lost in the cRBM, so one must utilize Metropolis-Hastings sampling with local update rules, which has significantly longer autocorrelation times, making it slower and more computationally expensive.

2. DBM wavefunctions

A possible quantum extension of DBMs based on the same principle of the cRBM was introduced in Ref. [9] and reads

$$\psi(\mathbf{x}; \boldsymbol{\theta}) = \sum_{\mathbf{x}^{(l)}_{l>0}} e^{\sum_{l=0}^{N_L-1} \mathbf{x}^{(l)T} W^{(l)} \mathbf{x}^{(l+1)} + \sum_{l=0}^{N_L} \mathbf{b}^{(l)T} \mathbf{x}^{(l)}}. \quad (\text{C3})$$

The original reference also provides an interesting constructive approach that does not require a variational optimization of the DBM parameters. However, due to the intractability of the summation in Eq. (C3) and the restrictive conditions under which efficient block Gibbs sampling of the model is possible, this variational Ansatz has only been applied to small-scale benchmarks.

3. Quantum Boltzmann Machines

Another possible extension, dubbed Quantum Boltzmann Machine (QBM), was proposed in Ref. [70]. The units of the BM are promoted to quantum spins, and the state of the system is described as the exponential of a transverse-field Ising Hamiltonian

$$H_{\text{QBM}} = - \sum_a \Gamma_a \sigma_a^x - \sum_a b_a \sigma_a^z - \sum_{ab} w_{ab} \sigma_a^z \sigma_b^z \quad (\text{C4})$$

yielding the density matrix operator

$$\rho = \frac{1}{Z} \exp\{-H_{\text{QBM}}\}. \quad (\text{C5})$$

The presence of non-commuting terms poses significant challenges in training the model, as detailed in the original reference [70], which applies this framework to address classical machine learning tasks. Later, Ref. [75] used a simplified version of QBMs without hidden units to learn the ground-states of quantum systems through a hybrid quantum-classical method inspired by the variational quantum eigensolver.

Appendix D: Analytical results for Boltzmann Quantum States

In this Section, we report additional results on BQS and their relationship to the cRBM introduced in [9] and described in Appendix C. We adopt the same notation as in Section VI.

1. Ground-state structure of the extended Hilbert space and expectation values

The BQS formalism describes both visible and hidden spins as quantum spins in an extended Hilbert space $\mathcal{H}_{vh} = \mathcal{H}_v \otimes \mathcal{H}_h$. Although the two Hamiltonians H_v and

$H_{vh} = H_v \otimes \mathbb{I}_h$ share the same spectrum, the ground-state of the extended system can have a non-trivial entanglement structure. Specifically, if H_v has a degenerate ground-state, then there exist ground-states of H_{vh} that are entangled across the visible/hidden partition, rather than simple tensor products of two pure states. Let H_v be a Hamiltonian with a d -fold degenerate ground-state, let $\{|\Psi_v^a\rangle\}_{a=1}^d$ be a set of orthonormal ground states, and let $\{|\Phi_v^b\rangle\}_{b=1}^{\dim \mathcal{H}_v - d}$ be a set of orthonormal higher energy eigenstates so that $\{|\Psi_v^a\rangle, |\Phi_v^b\rangle\}$ forms a complete

orthonormal basis of $\mathcal{H}_v$. Then, any state $|\psi_{vh}\rangle$ of the extended system can be expanded as

$$|\psi_{vh}(G, E)\rangle = \sum_{a, \mathbf{h}} G_{a\mathbf{h}} |\Psi_v^a\rangle \otimes |\mathbf{h}\rangle + \sum_{b, \mathbf{h}} E_{b\mathbf{h}} |\Phi_v^b\rangle \otimes |\mathbf{h}\rangle, \quad (\text{D1})$$

and its energy expectation value is

$$\frac{\langle \psi_{vh}(G, E) | H_{vh} | \psi_{vh}(G, E) \rangle}{\langle \psi_{vh}(G, E) | \psi_{vh}(G, E) \rangle} = \frac{\sum_{a, \mathbf{h}} |G_{a\mathbf{h}}|^2 \langle \Psi_v^a | H_v | \Psi_v^a \rangle + \sum_{b, \mathbf{h}} |E_{b\mathbf{h}}|^2 \langle \Phi_v^b | H_v | \Phi_v^b \rangle}{\sum_{a, \mathbf{h}} |G_{a\mathbf{h}}|^2 + \sum_{b, \mathbf{h}} |E_{b\mathbf{h}}|^2}. \quad (\text{D2})$$

Hence, by the variational principle, the ground states of H_{vh} take the form

$$|\Psi_{vh}(G)\rangle = \sum_{a, \mathbf{h}} G_{a\mathbf{h}} |\Psi_v^a\rangle \otimes |\mathbf{h}\rangle, \quad (\text{D3})$$

and variational energy minimization in the extended Hilbert space can generally yield entangled states that does not decompose as the tensor product of a state of the visible spins and a state of the hidden ones, regardless of the variational Ansatz used. Consequently, it is not generally possible to trace out the hidden degrees of

freedom to obtain a pure state for the visible spins. While it is theoretically possible to obtain a reduced density matrix for the visible spins to describe the mixed state in which they are found

$$\rho_v = \sum_{\mathbf{v}, \mathbf{v}', \mathbf{h}} \psi(\mathbf{v}, \mathbf{h}) \overline{\psi(\mathbf{v}', \mathbf{h})} |\mathbf{v}'\rangle \langle \mathbf{v}|, \quad (\text{D4})$$

the summation over $\mathbf{h}$ required to compute the matrix elements is tractable in the case of RBQS Ansätze, and an analytical expression can be obtained for the reduced density operator

$$\rho_v^{\text{RBQS}} = \sum_{\mathbf{v}, \mathbf{v}'} \exp \left\{ \left[\sum_i b_i v_i - \bar{b}_i v_i \right] \right\} \prod_j \cosh \left\{ \sum_i (W_{ij} v_i - \bar{W}_{ij} v'_i) + c_j - \bar{c}_j \right\} |\mathbf{v}'\rangle \langle \mathbf{v}|. \quad (\text{D5})$$

Similarly to the classical case, the summation is untractable for general BQS Ansätze, and it won't be possible to obtain a closed formula for the density operator.

It is however possible to obtain the ground-state expectation value of any operator O_v acting solely on visible spins from the ground state of the extended system by computing the expectation value of the operator $O_v \otimes \mathbb{I}_h$. Expectation values on a general ground state of the extended system are more conveniently computed using the Schmidt decomposition [?]: given any state of the form of Eq. (D3), let $(U, \tilde{G}, V)$ be the singular value decomposition of G , $G = U \tilde{G} V^\dagger$, then we can write $|\Psi\rangle_{vh}$ as

$$|\Psi_{vh}(G)\rangle = \sum_n \tilde{G}_n |\tilde{\Psi}_v^n\rangle \otimes |\tilde{\mathbf{h}}^n\rangle, \quad (\text{D6})$$

where $|\tilde{\Psi}_v^n\rangle = U |\Psi_v^a\rangle$, $|\tilde{\mathbf{h}}^n\rangle = V^\dagger |\mathbf{h}\rangle$, and $\tilde{G}_n = \tilde{G}_{nn}$. Using such expansion immediately expose that the operator $O_v \otimes \mathbb{I}_h$ on a general ground state of the extended system

is

$$\begin{aligned} & \frac{\langle \Psi_{vh}(G) | O_v \otimes \mathbb{I}_h | \Psi_{vh}(G) \rangle}{\langle \Psi_{vh}(G) | \Psi_{vh}(G) \rangle} \\ &= \frac{\sum_{n, m} \tilde{G}_n G_m \langle \tilde{\Psi}_v^n | O_v | \tilde{\Psi}_v^m \rangle \langle \tilde{\mathbf{h}}^n | \tilde{\mathbf{h}}^m \rangle}{\sum_{a, \mathbf{h}} |G_n|^2} \\ &= \sum_n p_n \langle \tilde{\Psi}_v^n | O_v | \tilde{\Psi}_v^n \rangle, \end{aligned} \quad (\text{D7})$$

where $p_n = |G_n|^2 / (\sum_m |G_m|^2)$. Unsurprisingly, this is the same result that would be found by first tracing out the hidden units to obtain a reduced density matrix for the visible units ρ_v , and then computing $\langle O_v \rangle = \text{tr}\{\rho_v O_v\}$.

In practical application, after a variational optimization, a DBQS converges to a close approximation of some ground state of the extended system $|\Psi_{vh}\rangle$. Then, this approximation can be used to efficiently sample the extended Hilbert space, and estimate the expectation value

of any operator of interest by tensoring it with the identity for the hidden units and computing the expectation value via Monte Carlo on the variational state.

2. Relation to the cRBM architecture and universality of RBQS

Here, we show that applying a projection operator to the hidden partition collapses the visible spins into a pure state. Then, we demonstrate that specific projections reduce the RBQS and DBQS Ansätze to the cRBM and DBM wavefunctions of Refs. [2, 9], establishing that the DBQS framework generalizes these earlier models.

Given a state of the hidden spins $|\chi_h\rangle$, applying the projection operator $P_\chi = \mathbb{I}_v \otimes |\chi_h\rangle\langle\chi_h|$ to a generic state of the extended system $|\psi_{vh}\rangle$ yields

$$\begin{aligned}
& (\mathbb{I}_v \otimes |\chi_h\rangle\langle\chi_h|) |\psi_{vh}\rangle \\
&= \sum_{\mathbf{v}, \mathbf{h}} |\mathbf{v}, \mathbf{h}\rangle \langle\mathbf{v}, \mathbf{h}| (\mathbb{I}_v \otimes |\chi_h\rangle\langle\chi_h|) \sum_{\mathbf{v}', \mathbf{h}'} |\mathbf{v}', \mathbf{h}'\rangle \langle\mathbf{v}', \mathbf{h}'| \psi_{vh}\rangle \\
&= \sum_{\mathbf{v}, \mathbf{h}, \mathbf{v}', \mathbf{h}'} |\mathbf{v}, \mathbf{h}\rangle \langle\mathbf{v}, \mathbf{h}'| \langle\mathbf{h}|\chi_h\rangle \langle\chi_h|\mathbf{h}'\rangle \langle\mathbf{v}', \mathbf{h}'| \psi_{vh}\rangle \\
&= \sum_{\mathbf{v}, \mathbf{h}, \mathbf{h}'} |\mathbf{v}, \mathbf{h}\rangle \langle\mathbf{h}|\chi_h\rangle \langle\chi_h|\mathbf{h}'\rangle \langle\mathbf{v}, \mathbf{h}'| \psi_{vh}\rangle \\
&= \sum_{\mathbf{v}} \left[\sum_{\mathbf{h}'} \langle\chi_h|\mathbf{h}'\rangle \langle\mathbf{v}, \mathbf{h}'| \psi_{vh}\rangle \right] |\mathbf{v}\rangle \otimes \sum_{\mathbf{h}} \langle\mathbf{h}|\chi_h\rangle |\mathbf{h}\rangle \\
&= \sum_{\mathbf{v}} \left[\sum_{\mathbf{h}'} \langle\chi_h|\mathbf{h}'\rangle \langle\mathbf{v}, \mathbf{h}'| \psi_{vh}\rangle \right] |\mathbf{v}\rangle \otimes |\chi_h\rangle.
\end{aligned} \tag{D8}$$

Now consider the special case where $|\psi_{vh}\rangle = |\Psi_{RBQS}\rangle$ is an (unnormalized) RBQS wavefunction as in Eq. (20), and

$$|\chi_h\rangle = |\rightarrow_h\rangle = \sum_{\mathbf{h}} |\mathbf{h}\rangle$$

is the (unnormalized) uniform superposition over all hidden configurations. Then, the projected state becomes

$$\begin{aligned}
& (\mathbb{I}_v \otimes |\rightarrow_h\rangle\langle\rightarrow_h|) |\Psi_{RBQS}\rangle \\
&= \sum_{\mathbf{v}} \left[\sum_{\mathbf{h}'} \exp\{\mathbf{v}^T W \mathbf{h}' + \mathbf{b}^T \mathbf{v} + \mathbf{c}^T \mathbf{h}'\} \right] |\mathbf{v}\rangle \otimes |\rightarrow_h\rangle \\
&= \sum_{\mathbf{v}} \left[e^{\sum_{i=0}^{N_v} b_i v_i} \prod_{j=0}^{N_h} \cosh\left\{c_j + \sum_{i=0}^{N_v} v_i W_{ij}\right\} \right] |\mathbf{v}\rangle \otimes |\rightarrow_h\rangle.
\end{aligned} \tag{D9}$$

Hence, the state in which the visible spins are found exactly matches the cRBM wavefunction introduced in Ref. [2]. Similarly, projecting a DBQS wavefunction with two hidden layers onto $|\rightarrow_h\rangle$ yields the DBM wavefunction of Ref. [9], where the visible amplitudes result from marginalizing over intermediate hidden layers. This establishes that both the cRBM and DBM can be viewed

as specific subclasses of the RBQS and DBQS Ansätze, respectively. Remarkably, since the DBM wavefunction family has been proved to be a universal approximator of quantum states [76], the DBQS inherits this expressive power, while potentially representing complex quantum states more effectively.

3. Dataset-free initialization of the DBQS parameters

Typical uses of DBMs in machine learning rely on greedy layer-wise pretraining [47]. In the NQA setting such pretraining yields no benefit: the first few optimization steps rapidly overwrite any pre-trained weights, effectively undoing the layer-wise initialization long before the algorithm approaches the hard part of the evolution. To our knowledge, however, dataset-free initialization of DBMs has not been systematically explored. Motivated by Ref. [77], which shows that Xavier initialization [78] improves training of canonical RBMs, we adopt a Xavier-normal initialization scheme for the variational parameters: all biases are set to zero, and the real and imaginary parts of the weights connecting layer (l) and layer ($l+1$) are sampled independently from a normal distribution with 0 mean and variance $a/(N_{(l)} + N_{(l+1)})$, where $N_{(l)}$ is the number of units in layer l and a is a gain factor (here fixed at $a = 10^{-2}$).

If a BQS parameters are drawn from $\mathcal{N}(0, \sigma^2)$, the ensemble-averaged log-amplitude $\mathbb{E}[\log \psi(\mathbf{x})]$ is exactly zero for any spin configuration $|\mathbf{x}\rangle$, and the variance scales linearly with σ . Hence, for sufficiently small gain factor the untrained network already provides a reasonable representation of the $|\rightarrow\rangle$ state.

4. Comparison of block Gibbs chain and MH chain

In Markov Chain Monte Carlo (MCMC) sampling, successive samples are typically correlated due to the Markovian nature of the process. This autocorrelation reduces the efficiency of the sampling procedure, as it decreases the number of effectively independent samples obtained from the chain. The degree of this dependence can be quantified using the integrated autocorrelation time (IAT), which provides a measure of the number of iterations required, on average, to produce an effectively independent sample.

Let $\{\mathbf{x}_t\}_{t=0}^{N-1}$ denote a stationary Markov chain, and consider a real-valued function of the chain state $f(\mathbf{x})$. The autocorrelation function at lag t is defined as

$$\rho_f(t) = \frac{\mathbb{E}[(f(\mathbf{x}_0) - \mu)(f(\mathbf{x}_t) - \mu)]}{\sigma} \tag{D10}$$

where μ and σ denote the mean and variance of $f(x)$ on

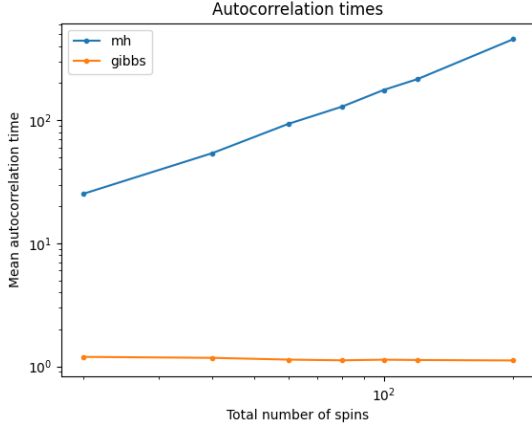


Figure 6. Comparison of the magnetization IAT between the single spin flip Metropolis-Hasting chain and the block Gibbs chain.

the chain. The IAT can be estimated by

$$\tau_f = 1 + 2 \sum_{t=1}^{+\infty} \rho_f(t) \quad (\text{D11})$$

In practice, the infinite sum is truncated at a lag where $\rho_f(t)$ becomes negligible. When considering multiple functions, one should consider the highest IAT across all functions to ensure reliable statistical estimates. In the context of spin systems, where the state is represented by spin variables, the function of interest are typically the individual spins: $f_i(\mathbf{x}) = x_i$. For MCMC algorithms employing local update rules, such as the single-spin flip Metropolis-Hasting chain, the autocorrelation time often scales at least linearly with the system size. Specifically, if only a fraction α of the N_s spins are updated in each iteration, at least N_s/α updates are required to provide each spin an opportunity to change, thereby influencing the overall autocorrelation time. Moreover, Metropolis-Hasting typically has an acceptance rate about 25% further increasing the autocorrelation times, as opposed to the Gibbs chain that accepts every update automatically.

We run both Gibbs chains and MH chains for $N = 10$ realizations of DBQS with different system sizes, 3 layers and hidden units density per layer of 1. figure 6 shows the average IAT for untrained DBQS with 3 hidden layers and hidden units density per layer of 1 for different system sizes. As expected, we find that the block Gibbs chain systematically converge faster and have autocorrelation times of order $O(1)$ compared to the single spin flip metropolis hasting chain that has IAT of order $O(N)$.

Appendix E: Algorithmic details on Neural Quantum Annealing

This section provides supplementary technical details on NQA. In particular, we specify the exact implemen-

tation of our code and list all tunable hyperparameters, commenting on their effect on algorithmic behavior and computational cost. From a broad algorithmic perspective, NQA is an advanced Monte Carlo method: its runtime scales polynomially, and it does not guarantee finding the exact solution in a strict mathematical sense. Despite the lack of performance guarantees, for many spin glass problems described in Section VIII, it successfully finds the best or best-known solution, or a good approximation. As for any Monte Carlo algorithm, the quality of this approximation can be systematically improved through careful hyperparameter tuning.

1. Algorithmic overview

As discussed in Section VII, NQA combines standard SR with momentum acceleration and persistent Gibbs chains enabled by our novel DBQS architecture. The update step is shown in Algorithm 3, and recursively executed throughout the optimization. It comprises three actions:

1. **Sampler update:** Advance the persistent chains with a few block-Gibbs updates and draw a set of configurations;
2. **Natural-gradient evaluation:** Compute the natural gradient;
3. **Parameter update:** Update the variational parameters and the optimizer state.

This update step is embedded into the full NQA optimization showcased in Algorithm 4, which comprises the following actions:

1. **Initialization:** The parameters of the DBQS are drawn from the Xavier-normal distribution as described in Section D3. An ensemble of block-Gibbs chains is then initialized and thermalized.
2. **Pre-training:** The Ansatz is first trained to approximate the ground state of the initial Hamiltonian at $s_0 = 0$. When the initial Hamiltonian is chosen as $H_0 = -\sum_i \sigma_i^x$, this step can be skipped because, as argued in Section D3, the Xavier initialization already yields an accurate approximation to its ground state.
3. **Iterative annealing:** For each intermediate annealing point s_j , we perform a few NQA update steps. In practice, a single step for each value of s_j ($j = 1, \dots, N_A - 1$) is usually sufficient to keep the state close to the variational adiabatic path and successfully approximate the final target state.
4. **Fine-tuning:** Finally, additional update steps are carried out at $s_{N_A} = 1$ in order to suppress any residual excited-state components of the wavefunction.

Algorithm 3 NQA_UpdateStep

Require: Variational parameters θ ; Persistent Gibbs chains $\mathcal{C}$; Optimizer state α .
Ensure: Updated parameters θ ; Updated Gibbs chains $\mathcal{C}$; Updated optimizer state α .
 1: Update the Gibbs chains $\mathcal{C}$ by applying N_{sweep} block Gibbs updates, and sample a set of configurations
 2: Compute the natural gradient $\mathbf{g} \leftarrow \mathbf{g}_\theta E(\theta)$
 3: Compute the updates for the parameters and the new optimizer state $(\delta\theta, \alpha) \leftarrow \text{opt}(\mathbf{g}, \alpha)$
 4: Update the variational parameters $\theta \leftarrow \theta + \delta\theta$
 5: **return** $\theta, \mathcal{C}, \alpha$

Algorithm 4 NQA for ground-state search

Require: Number of warm-up steps N_W ; Number of annealing steps N_A ; Number of updates per annealing step N_U ; Number of fine-tuning steps N_F ; Optimizer $\text{opt}(\mathbf{g}, \alpha)$ (natural gradients with momentum); Variational Ansatz $\psi(\mathbf{x}; \theta)$ (DBQS).
Ensure: Optimized parameters θ
 1: Initialize parameters $\theta \leftarrow \theta_0$ ▷ Initialization
 2: Initialize auxiliary state of the optimizer $\alpha \leftarrow \alpha_0$
 3: Initialize and thermalize Gibbs Chains $\mathcal{C}$
 4: $H(s) \leftarrow H_0$
 5: **for** $n = 0$ **to** $N_W - 1$ **do** ▷ Pretraining
 6: $\theta, \mathcal{C}, \alpha \leftarrow \text{NQA_UpdateStep}(\theta, \mathcal{C}, \alpha)$
 7: **end for**
 8: **for** $j = 1$ **to** $N_A - 1$ **do** ▷ Iterative annealing
 9: $s \leftarrow j/N_A$
 10: $H(s) \leftarrow sH_T + (1-s)AH_0 + s(1-s)BH_C$
 11: **for** $n = 0$ **to** $N_U - 1$ **do**
 12: $\theta, \mathcal{C}, \alpha \leftarrow \text{NQA_UpdateStep}(\theta, \mathcal{C}, \alpha)$
 13: **end for**
 14: **end for**
 15: $H(s) \leftarrow H_T$
 16: **for** $n = 0$ **to** $N_F - 1$ **do** ▷ Finetuning
 17: $\theta, \mathcal{C}, \alpha \leftarrow \text{NQA_UpdateStep}(\theta, \mathcal{C}, \alpha)$
 18: **end for**
 19: **return** θ

2. NQA Hyperparameters

In the following, we enumerate the tunable hyperparameters of NQA, organizing them into five groups based on their role in the algorithm.

a. Annealing Hyperparameters These parameters characterize the annealing schedule and the parametric Hamiltonian $H(s)$ in Eq. (32).

- Number of annealing steps N_A : it fixes the number of discretized values of the annealing parameter along the schedule, precisely given by $0 = s_0 < s_1 < \dots < s_{N_A} = 1$.
- Number of warm-up update steps N_W : number of `NQA_UpdateSteps` to perform for $s_0 = 0$ to prepare the initial ground-state.
- Number of updates steps per annealing step N_U : number of `NQA_UpdateSteps` performed for each intermediate value of s_j , for $j = 1, \dots, N_A - 1$. Using

a number of updates per step equal to one is sufficient for most classical and quantum applications, by selecting a sufficiently large number of annealing steps N_A , as anticipated in Section VII.

- Number of fine-tuning update steps N_F : number of `NQA_UpdateSteps` performed at $s_{N_A} = 1$. Whereas this final refinement does not seem to yield any performance improvement on classical problems, it empirically improves the performance for quantum ground states.
- Annealing field scale $A > 0$: multiplicative prefactor of the initial Hamiltonian H_0 . Typically, during the annealing dynamics, the system undergoes a phase transition (a crossover, for finite systems) from a paramagnetic to a spin glass phase (additional phase transitions, such as perturbative crossings, may also occur). Heuristically, NQA is expected to work better when such a transition occurs somewhat close to $s \approx 1/2$. Indeed, if the transition happens for $s \approx 0$, there is almost no difference between NQA and standard NQS optimization on the target problem Hamiltonian, since the hard regime is immediately reached. On the contrary, when the transition happens for $s \approx 1$, the variational Ansatz may not have enough time to relax on the target ground-state in the hard regime, and computational resources may be wasted in the easy regime. Correctly tuning A such that the transition occurs neither too soon nor too late during the annealing can be achieved automatically through hyperparameter optimization.
- Catalyst field scale B : multiplicative prefactor of H_C , included for the same reason as the annealing field scale. If we explicitly fix $B = 0$, no catalyst is utilized.

b. Optimizer Hyperparameters As stated in the main text, we use natural gradients with momentum. Hence, the optimizer hyperparameters are the learning rate $\eta > 0$, momentum $0 \leq \mu < 1$, and diagonal shift λ_d . The diagonal shift does not appear to be critical for the algorithm's performance. On the contrary, selecting optimal learning rate and momentum values is crucial and should be done automatically via hyperparameter optimization, as the optimal values strongly depend on the target problem and on the NQS architecture.

c. DBQS Hyperparameters

- Number of hidden layers N_L : number of hidden layers in the DBQS architecture defined in Section VI.
- Units density per layer α : ratio between the number of hidden spins in each hidden layer (assumed to be layer-independent) and the number of visible (physical) spins in the visible layer.
- Parameters type (real or complex): whether to use real or complex variational parameters θ . Real parameters can be used for stoquastic Hamiltonians

as the instantaneous ground state can be represented by real and positive amplitudes. Using real parameters enforces this inductive bias, reduces the capacity of the Ansatz, and may yield reduced computational cost and runtime.

d. Block Gibbs sampler Hyperparameters

- Number of samples N_{samples} : number of samples to generate to compute observables, such as the local energy.
- Number of chains N_{chains} : number of persistent block Gibbs chains to run in parallel. We recommend having one chain per sample, if possible, as this yields a faster runtime and guarantees that the samples are uncorrelated.
- Number of thermalization steps N_{therm} : number of Gibbs chain steps to thermalize the chains at the beginning of the algorithm. (Recommended values may be 100 – 1000).
- Number of sweep updates N_{sweeps} : number of block Gibbs updates to perform in between drawing successive samples from a given chain. (Recommended 10 – 100).

3. Hyperparameters tuning

Finding the optimal hyperparameters for the algorithms is challenging as they correlate in a non trivial manner. The optimal choice depends from the problem at hand, and as for most of machine learning algorithm there is no universal set of hyperparameters that works every time for every problem. To find the optimal hyperparameters for a given problem we use Optuna [64], a framework for black box optimization commonly used in the machine learning community for hyperparameters tuning. The specific algorithms we use are the Tree-structured Parzen Estimator (TPE) [79] and the Covariance Matrix Adaptation Evolution Strategy (CMA) [80] depending on the computational budget and the parallelization capabilities available.

a. TPE in short: on each trial and for each parameter fits two Gaussian Mixture Models $l(x)$ and $g(x)$ the first one to the set of values associated to the best objective value and the second to the remaining values. Then, it chooses the value that maximizes $l(x)/g(x)$.

b. CMA-ES in short: it is an evolutionary algorithm that operates by evolving a population of candidate solutions. It samples a batch of hyperparameter sets from a multivariate normal distribution, evaluates each and then updates the distribution’s parameters based on the top-performing samples. However, it typically require a large number of evaluations to reach its potential.

HPO Space

Here we report the hyperparameter ranges for HPO for the classical and quantum benchmark discussed in the main text. We observe that the final finetuning stage can improve performance as measured by the final energy of the variational state, but never improves the performance as measured by the best classical configuration encountered. As discussed in the main text, for classical problems the success criterion can be relaxed to encountering a ground-state configuration during the simulation, therefore an aggressive schedule can be used.

For classical problems we tune the following parameters in the corresponding ranges, hyperparameters that range across multiple orders of magnitudes are sampled in log-space

- Number of annealing steps (int, log): [1000, 1000000],
- Annealing field scale (float, log): [0.1, 10],
- Catalyst field scale (float, log): [0.1, 10] (or set 0 if real parameters),
- Number of hidden layers (int, linear): [1, 3],
- Unit density per layer (float, log): [0.25, 4],
- Number samples (int, log): [8, 128],
- Number sweep steps (int, log): [4, 128],
- Learning rate (float, log): [0.0001, 1],
- Momentum (float, linear): [0.0, 0.9],
- Diagonal shift (float, log): [1e-9, 1e0],

while fixing

- Number of warm-up update steps: 1,
- Number of update steps per annealing steps: 1,
- Number of finetuning steps: 1,
- Number of chains equal to the number of samples,
- Number of thermalization steps 128.

And, for quantum problems we tune

- Unit density per layer (float, log): [0.25, 4],
- Number of annealing steps (int, log): [1000, 1000000],
- Number sweep steps (int, log): [4, 128],
- Learning rate (float, log): [0.0001, 1],
- Momentum (float, linear): [0.0, 0.9],

while fixing

- Number of warm-up update steps: 10,
- Number of update steps per annealing steps: 1,
- Number of finetuning steps: 100,
- Number of samples 201,
- Number of chains equal 201,
- Number of thermalization steps 128,
- Annealing field scale: 1,
- Catalyst field scale: 1,
- diagonal shift 0.001.

4. Computational complexity

The computational complexity of NQA is linear in the number of calls of the `NQA.UpdateStep` function, which amounts to $N_W + N_F + N_A N_U$. The first two terms in the sum are usually negligible, as is the computational cost of the rest of the code; hence, the overall computational cost is essentially proportional to $N_A N_U$ times the computational cost of a `NQA.UpdateStep` function call. The latter comprises three contributions; we will now analyze them one by one.

First, propagating each persistent block Gibbs chain for N_{sweeps} sweeps through a DBQS with N_L hidden layers requires $O(N_{\text{sweeps}} N_L)$ matrix multiplications of matrices of size $O(N_v)$ times a vector of size $O(N_v)$. An operation that has computational complexity of $O(N_v^2)$, for a total complexity of $O(N_{\text{sweeps}} N_L N_v^2)$, and updating all the chains costs $O(N_{\text{samples}} N_{\text{sweeps}} N_L N_v^2)$.

Second, the complexity of computing the natural gradients via SR scales as $O(N_{\text{params}}^3)$ or via min-SR $O(N_{\text{params}} N_{\text{samples}}^2 + N_{\text{samples}}^3)$ [7] for the matrix pseudoinverse. Since a DBQS has number of parameters $O(N_{\text{spins}}^2 N_{\text{layers}})$, the SR route costs $O(N_{\text{spins}}^6 N_{\text{layers}}^3)$, and min-SR reduces this to $O(N_{\text{spins}}^2 N_{\text{layers}} N_{\text{samples}}^2 + N_{\text{samples}}^3)$. There is also the calculation of the logarithmic derivatives that is just computing averages from the samples previously generated $O(N_{\text{params}} N_{\text{samples}})$.

And the local energies, that as mentioned in the main text they involve a matrix multiplication $O(N_v^2)$ and a scalar product $O(N_v)$. Total cost via min-SR is $O(N_{\text{params}} N_{\text{samples}}^2 + N_{\text{samples}}^3)$.

Third, evaluating the local energy and related observables is $O(N_{\text{spins}}^2)$, while the subsequent momentum-SGD update of the parameters is negligible.

Collecting all terms, the per-step complexity for the practical (min-SR) implementation is

$$O(N_{\text{spins}}^2 N_{\text{layers}} (N_{\text{samples}}^2 + N_{\text{sweeps}}) + N_{\text{samples}}^3), \quad (\text{E1})$$

which remains tractable for the modest $N_{\text{samples}}, N_{\text{sweeps}} \ll N_{\text{spins}}^2$ typically required in practice.

Appendix F: Additional information on the classical benchmarks

Table I reports the typical energy errors on the $N = 200$ benchmarks for all the combinations explored. Figure 7 and Figure 8 show the energy error histograms for the set-ups not shown in the main text.

Method	Typical final/best energy error
cRBM-SR	1.360e-01/1.353e-01
cRBM-SR-cata	9.685e-02/9.661e-02
cRBM-NQA	1.687e-01/1.653e-01
cRBM-NQA-cata	3.280e-01/3.207e-01
RBQS-SR	1.347e-02/1.311e-02
RBQS-SR-cata	8.510e-03/8.510e-03
RBQS-NQA	2.274e-06/6.107e-07
RBQS-NQA-cata	3.872e-06/7.614e-08
DBQS-NQA-cata	7.883e-07/1.256e-08

Table I. Typical energy errors on the $N = 200$ spins SK benchmarks by method.

Appendix G: Transverse field SK results

In the context of TFSK model, we plot the relative energy (given by Target energy - lowest energy) vs variance highlighting the low lying excited states and the energy vs number of parameters for the same 10 instances of 100 spins. All of the figures below are for a transverse field value of $g = 0.1$.

-
- [1] D-Wave Systems Inc., *Advantage Processor Overview*, Technical Report 14-1058A (D-Wave Systems, 2020) accessed: yyyy-mm-dd.
 - [2] G. Carleo and M. Troyer, *Science* **355**, 602–606 (2017).
 - [3] M. Hibat-Allah, E. M. Inack, R. Wiersema, R. G. Melko, and J. Carrasquilla, *Nature Machine Intelligence* **3**, 952–961 (2021).
 - [4] <http://spinglass.uni-bonn.de/>.
 - [5] D.-L. Deng, X. Li, and S. Das Sarma, *Phys. Rev. X* **7**, 021021 (2017).
 - [6] C. Roth, A. Szabó, and A. H. MacDonald, *Phys. Rev. B* **108**, 054410 (2023).
 - [7] A. Chen and M. Heyl, *Nature Physics* **20**, 1476 (2024).
 - [8] A. Chen, V. D. Naik, and M. Heyl, *Convolutional transformer wave functions* (2025), [arXiv:2503.10462](https://arxiv.org/abs/2503.10462) [cond-mat.dis-nn].
 - [9] G. Carleo, Y. Nomura, and M. Imada, *Nature Communications* **9**, 10.1038/s41467-018-07520-3 (2018).
 - [10] M. Schmitt and M. Heyl, *Phys. Rev. Lett.* **125**, 100503 (2020).

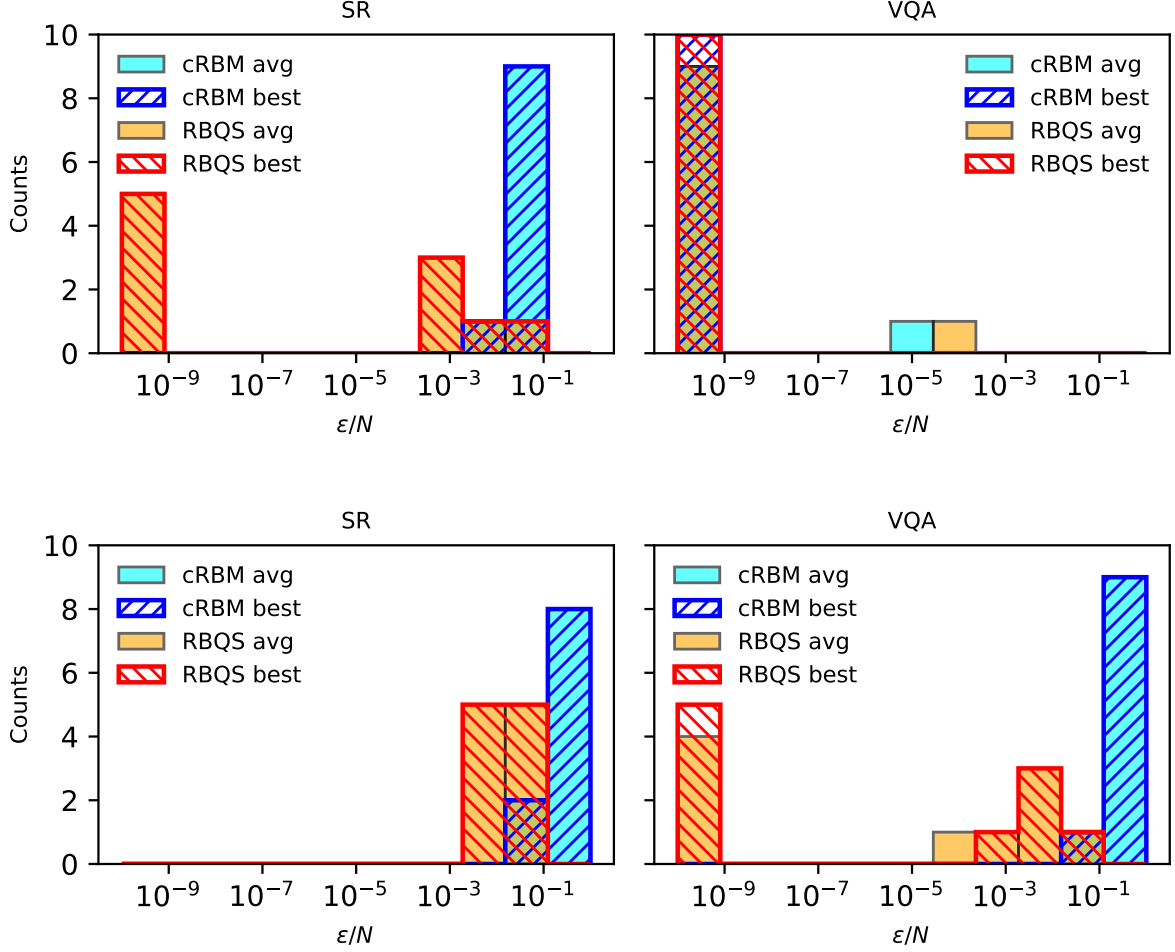


Figure 7. Residual energy error histograms for each combination of variational Ansatz and optimization strategy on the benchmark instances with $N = 100$ (top panels) and $N = 200$ (bottom panels) spins. Left panels show the comparison of the final energies of real parameterized cRBM and RBQS Ansatzes when optimized via SR. Right panels showcases the same comparison when the Ansatzes are optimized via VQA. Solid bars represent the average energy of the final quantum state, and dashed bars represent the lowest classical energy encountered during the simulation. Leftmost bars corresponds to instances solved exactly.

- [11] M. Schmitt, M. M. Rams, J. Dziarmaga, M. Heyl, and W. H. Zurek, *Science Advances* **8**, eabl6850 (2022), <https://www.science.org/doi/pdf/10.1126/sciadv.abl6850>.
- [12] T. Mendes-Santos, M. Schmitt, A. Angelone, A. Rodriguez, P. Scholl, H. J. Williams, D. Barredo, T. Lahaye, A. Browaeys, M. Heyl, and M. Dalmonte, *Phys. Rev. X* **14**, 021029 (2024).
- [13] H. Lange, A. V. de Walle, A. Abedinnia, and A. Bohrdt, *From architectures to applications: A review of neural quantum states* (2024), [arXiv:2402.09402 \[cond-mat.dis-nn\]](https://arxiv.org/abs/2402.09402).
- [14] M. Schmitt and M. Heyl, arXiv preprint [arXiv:2506.03124](https://arxiv.org/abs/2506.03124) (2025).
- [15] M. Rattray, D. Saad, and S.-i. Amari, *Physical review letters* **81**, 5461 (1998).
- [16] R. Pascanu and Y. Bengio, *Revisiting natural gradient for deep networks* (2014), [arXiv:1301.3584 \[cs.LG\]](https://arxiv.org/abs/1301.3584).
- [17] S. Sorella, *Physical review letters* **80**, 4558 (1998).
- [18] A. Ben-Israel and T. N. Greville, *Generalized inverses: theory and applications* (Springer Science & Business Media, 2006).
- [19] B. Polyak, *USSR Computational Mathematics and Mathematical Physics* **4**, 1 (1964).
- [20] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, *nature* **323**, 533 (1986).
- [21] I. Sutskever, J. Martens, G. Dahl, and G. Hinton, in *Proceedings of the 30th International Conference on International Conference on Machine Learning - Volume 28*, ICML'13 (JMLR.org, 2013) p. III-1139-III-1147.
- [22] A. Das and B. K. Chakrabarti, *Rev. Mod. Phys.* **80**, 1061 (2008).
- [23] A. B. Finnila, M. A. Gomez, C. Sebenik, C. Stenson, and J. D. Doll, *Chem. Phys. Lett.* **219**, 343 (1994).
- [24] T. Kadowaki and H. Nishimori, *Phys. Rev. E* **58**, 5355 (1998).

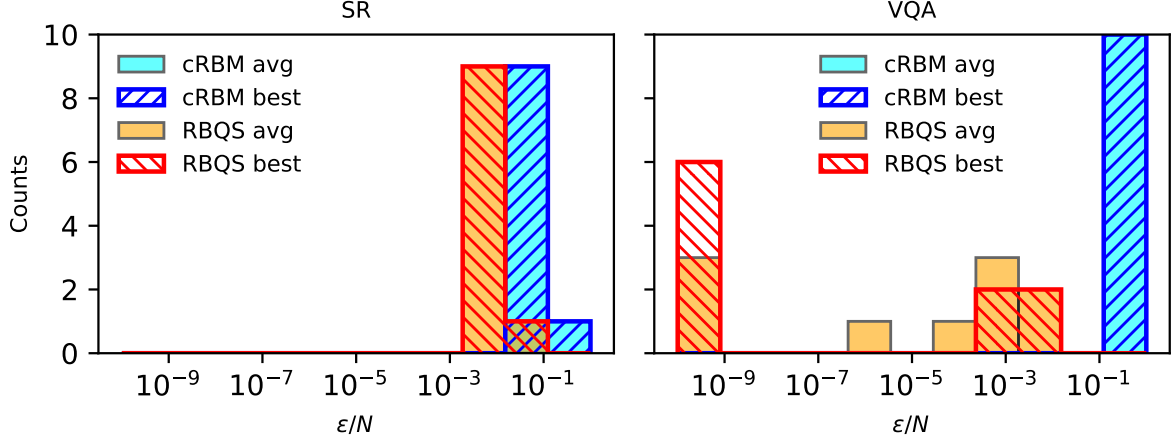


Figure 8. Residual energies histogram for each combination of variational Ansatz and optimization strategy for the 10 realizations of the SK model with $N = 200$ spins. Left panel shows the comparison of the final energies of complex parameterized cRBM and RBQS Ansätze when optimized via SR. Right panel showcases the same comparison when the Ansätze are optimized via NQA, with the inclusion of a catalyst. In both panels, the solid bars represent the average energy of the final quantum state, and the dashed bars represent the lowest classical energy encountered during the simulation. Leftmost bars corresponds to instances solved exactly.

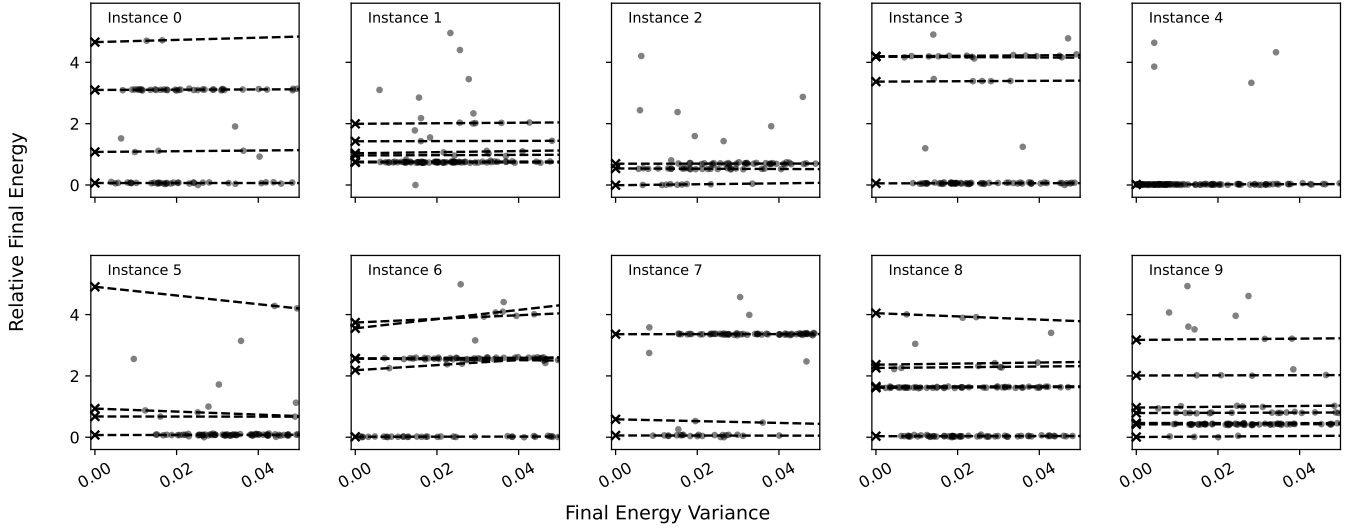


Figure 9. Relative final energy vs variance for all the instances. To produce the above plots, the energy ranges were spliced into small intervals and the extrapolation of energy levels were done to find the zero variance true energy for ground state and low lying excited eigenstates.

- [25] G. E. Santoro and E. Tosatti, *J. Phys. A: Math. Gen.* **39**, R393 (2006).
- [26] E. Farhi, J. Goldstone, S. Gutmann, J. Lapan, A. Lundgren, and D. Preda, *Science* **292**, 472 (2001).
- [27] T. Albash and D. A. Lidar, *Rev. Mod. Phys.* **90**, 015002 (2018).
- [28] F. Glover, G. Kochenberger, and Y. Du, A tutorial on formulating and using qubo models (2019), [arXiv:1811.11538 \[cs.DS\]](https://arxiv.org/abs/1811.11538).
- [29] A. Lucas, *Frontiers in Physics* **2**, 5 (2014).
- [30] M. H. S. Amin and V. Choi, *Physical Review A* **80**, 10.1103/physreva.80.062326 (2009).
- [31] I. Hen and A. P. Young, *Phys. Rev. E* **84**, 061152 (2011).
- [32] B. Altshuler, H. Krovi, and J. Roland, *Proceedings of the National Academy of Sciences* **107**, 12446 (2010), <https://www.pnas.org/doi/pdf/10.1073/pnas.1002116107>.
- [33] S. Knysh, *Nature Communications* **7**, 10.1038/ncomms12370 (2016).
- [34] S. Yarkoni, E. Raponi, T. Bäck, and S. Schmitt, *Reports on Progress in Physics* **85**, 104001 (2022).

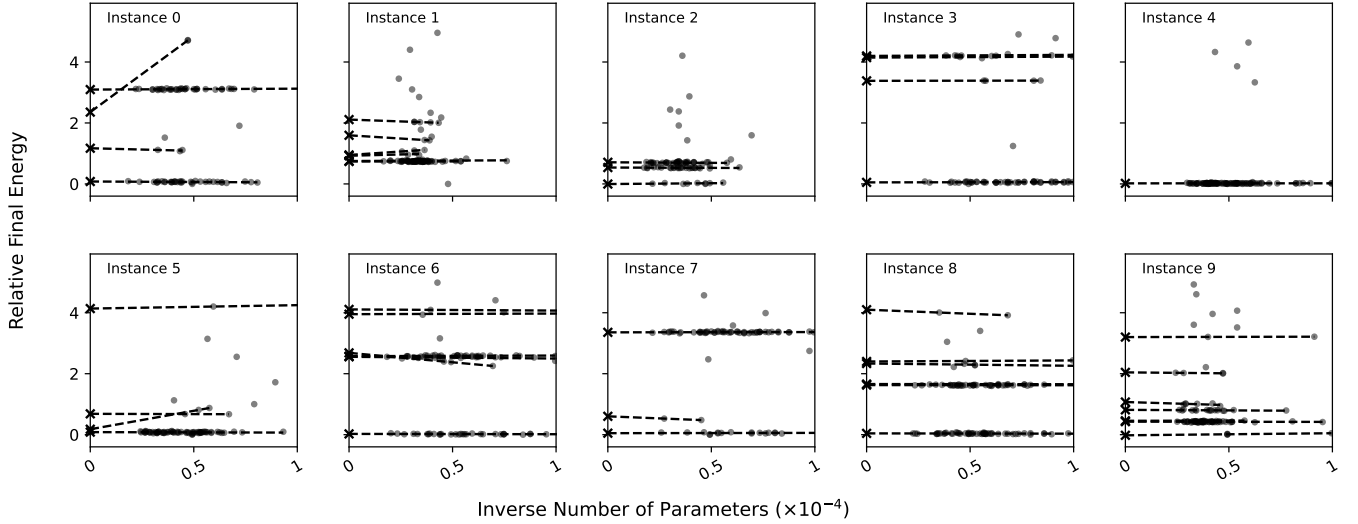


Figure 10. Relative final energy dependence on inverse number of parameters of DBQS ansatz. The infinite parameter extrapolated Energy is shown with 'X'. For consistency with the variance plots only points which have variance lower than 0.05 were taken into consideration.

- [35] J. Cai, W. G. Macready, and A. Roy, [A practical heuristic for finding graph minors](#) (2014), [arXiv:1406.2741 \[quant-ph\]](#).
- [36] P. Torta, R. Casati, S. Bruni, A. Mandarino, and E. Prati, *EPJ Quantum Technology* **12**, 66 (2025).
- [37] S. Schulz, D. Willsch, and K. Michielsen, [Learning-driven annealing with adaptive hamiltonian modification for solving large-scale problems on quantum devices](#) (2025), [arXiv:2502.21246 \[quant-ph\]](#).
- [38] T. F. Rønnow, Z. Wang, J. Job, S. Boixo, S. V. Isakov, D. Wecker, J. M. Martinis, D. A. Lidar, and M. Troyer, *Science* **345**, 420 (2014).
- [39] A. King, A. Nocera, M. Rams, J. Dziarmaga, A. Zucca, T. Lanting, J. Raymond, and et al., *Science* **388**, eado6285 (2025), published online March12,2025.
- [40] J. Tindall, A. Mello, M. Fishman, M. Stoudenmire, and D. Sels, [Dynamics of disordered quantum systems with two- and three-dimensional tensor networks](#) (2025), [arXiv:2503.05693 \[quant-ph\]](#).
- [41] L. Maunon and G. Carleo, [Challenging the quantum advantage frontier with large-scale classical simulations of annealing dynamics](#) (2025), [arXiv:2503.08247 \[quant-ph\]](#).
- [42] S. Ebadi, A. Keesling, M. Cain, T. T. Wang, H. Levine, D. Bluvstein, G. Semeghini, A. Omran, J.-G. Liu, M. D. Lukin, et al., *Science* **376**, 1209 (2022).
- [43] G. Lami, P. Torta, G. E. Santoro, and M. Collura, *SciPost Phys.* **14**, 117 (2023).
- [44] P. Torta, L. Leone, R. Casati, and E. Prati, in *2024 IEEE International Conference on Quantum Computing and Engineering (QCE)*, Vol. 02 (2024) pp. 514–515.
- [45] B. Žunkovič, P. Torta, G. Pecci, G. Lami, and M. Collura, *Phys. Rev. Lett.* **134**, 130601 (2025).
- [46] U. Schollwöck, *Annals of Physics* **326**, 96 (2011), january 2011 Special Issue.
- [47] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning* (MIT Press, 2016) <http://www.deeplearningbook.org>.
- [48] N. Le Roux and Y. Bengio, *Neural Computation* **20**, 1631 (2008), <https://direct.mit.edu/neco/article-pdf/20/6/1631/817339/neco.2008.04-07-510.pdf>.
- [49] N. Srivastava and R. R. Salakhutdinov, *Advances in neural information processing systems* **25** (2012).
- [50] R. D. Hjelm, V. D. Calhoun, R. Salakhutdinov, E. A. Allen, T. Adali, and S. M. Plis, *NeuroImage* **96**, 245 (2014).
- [51] F. He, X. Huang, X. Wang, S. Qiu, F. Jiang, and S. H. Ling, *Computerized Medical Imaging and Graphics* **89**, 101871 (2021).
- [52] E. Lorch, in *ICML Workshop on Visualization for Deep Learning* (2016).
- [53] I. Jolliffe, in *International encyclopedia of statistical science* (Springer, 2011) pp. 1094–1096.
- [54] M. Mezard, G. Parisi, M. Virasoro, and J. Hopfield, *Spin Glass Theory And Beyond: An Introduction To The Replica Method And Its Applications*, World Scientific Lecture Notes In Physics (World Scientific Publishing Company, 1987).
- [55] K. Hukushima and K. Nemoto, *Journal of the Physical Society of Japan* **65**, 1604–1608 (1996).
- [56] G. E. Santoro, R. Martoňák, E. Tosatti, and R. Car, *Science* **295**, 2427 (2002).
- [57] B. Heim, T. F. Rønnow, S. V. Isakov, and M. Troyer, *Science* **348**, 215–217 (2015).
- [58] T. Zanca and G. E. Santoro, *Phys. Rev. B* **93**, 224431 (2016).
- [59] C. Baldassi and R. Zecchina, *Proceedings of the National Academy of Sciences* **115**, 1457 (2018).
- [60] G. Goh, *Distill* **10.23915/distill.00006** (2017).
- [61] H. Nishimori and K. Takada, *Frontiers in ICT* **4**, 10.3389/fict.2017.00002 (2017).
- [62] R. Ghosh, L. A. Nutricati, N. Feinstein, P. A. Warburton, and S. Bose, [Exponential speed-up of quantum annealing via n-local catalysts](#) (2024), [arXiv:2409.13029 \[quant-ph\]](#).
- [63] K. Nishimura, Y. Sakamoto, T. Shimizu, K. Suzuki, and Y. Yamashiro, *OpenJij*.

- [64] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (2019).
- [65] M. E. Khan, D. Nielsen, V. Tangkaratt, W. Lin, Y. Gal, and A. Srivastava, [Fast and scalable bayesian deep learning by weight-perturbation in adam](#) (2018), [arXiv:1806.04854 \[stat.ML\]](#).
- [66] J. Martens and R. Grosse, [Optimizing neural networks with kronecker-factored approximate curvature](#) (2020), [arXiv:1503.05671 \[cs.LG\]](#).
- [67] C. Li, S. Zhu, Z. Xie, and Z. Wen, [Accelerated natural gradient method for parametric manifold optimization](#) (2025), [arXiv:2504.05753 \[math.OC\]](#).
- [68] P. M. Long and R. Servedio, in *Proceedings of the 27th International Conference on Machine Learning (ICML-10)* (Citeseer, 2010) pp. 703–710.
- [69] G. Casella and E. I. G. and, [The American Statistician](#) **46**, 167 (1992), <https://www.tandfonline.com/doi/pdf/10.1080/00031305.1992.10475858>.
- [70] M. H. Amin, E. Andriyash, J. Rolfe, B. Kulchytskyy, and R. Melko, *Physical Review X* **8**, [10.1103/physrevx.8.021050](#) (2018).
- [71] G. Torlai, G. Mazzola, J. Carrasquilla, M. Troyer, R. Melko, and G. Carleo, *Nature physics* **14**, 447 (2018).
- [72] W. Zhang, X. Xu, Z. Wu, V. Balachandran, and D. Poletti, *Phys. Rev. B* **107**, [165149](#) (2023).
- [73] L. L. Viteritti, F. Ferrari, and F. Becca, *SciPost Phys.* **12**, 166 (2022).
- [74] Y. Nomura, *Journal of Physics: Condensed Matter* **33**, [174003](#) (2021).
- [75] D. Patel, D. Koch, S. Patel, and M. M. Wilde, [Quantum boltzmann machine learning of ground-state energies](#) (2024), [arXiv:2410.12935 \[quant-ph\]](#).
- [76] X. Gao and L.-M. Duan, *Nature communications* **8**, 662 (2017).
- [77] M. Yasuda, R. Maeno, and C. Takahashi, *Neural Networks* **187**, 107297 (2025).
- [78] X. Glorot and Y. Bengio, in *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, Proceedings of Machine Learning Research, Vol. 9, edited by Y. W. Teh and M. Titterton (PMLR, Chia Laguna Resort, Sardinia, Italy, 2010) pp. 249–256.
- [79] J. Bergstra, R. Bardenet, Y. Bengio, and B. Kégl, *Advances in neural information processing systems* **24** (2011).
- [80] N. Hansen, *arXiv preprint arXiv:1604.00772* (2016).